EXERCISE SESSION 2 - GROUP 4

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EXERCISE 1

a) let the Flag be the random variable which measures whether a client is "Good" or "Bad" (with the usual value 1 associated to the level "Good"), X1 the binary random variable which measures the the marital status (Marital status taking value 0 if never married, 1 otherwise) and X2 the binary random variable which measures the home's ownership (Owner). Choose level "N" as a reference level for the variable Owner and determine the value of α, βX1, βX2,βX1X2;

In this case, we are asked to find the parameters of a model, such as alpha, Beta x1, Beta x2, and the interaction between x1 and x2. To find the values of the parameters, it will be necessary to use the logistic model, that is, the logistic regression model:

$$logit\,(P(Flag = 1 \mid X_1, X_2)) = \alpha + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \beta_{12} \cdot X_1 \cdot X_2$$

We can start building a theoretical dataframe.

```
df <- data.frame(
  Flag = c(0, 1, 0, 1, 0, 1, 0, 1),
  X1 = c(0, 0, 1, 1, 0, 0, 1, 1),  # Marital status
  X2 = c(0, 0, 0, 0, 1, 1, 1, 1),  # Ownership
  Prob = c(0.13, 0.22, 0.21, 0.12, 0.05, 0.08, 0.04, 0.15)
)
df
```

```
##   Flag X1 X2 Prob
## 1    0  0  0 0.13
## 2    1  0  0 0.22
## 3    0  1  0 0.21
## 4    1  1  0 0.12
## 5    0  0  1 0.05
## 6    1  0  1 0.08
## 7    0  1  1 0.04
## 8    1  1  1 0.15
```

Since we had theoretical distribution, we have to expand the data into observational form by multiplying the probabilities by 1.000 to obtain whole-number frequencies.

```
df$Freq <- round(df$Prob * 1000)
```

Creation of the expanded dataset.

```
df_expanded <- df[rep(1:nrow(df), df$Freq), ]
df_expanded$Flag <- as.factor(df_expanded$Flag)
```

To examine the main coefficients, we have two approaches: a faster method using logistic regression with interaction terms, or a more theoretical method based on calculating the log-odds.

We can start with the logistic regression with interaction (βX1X2).

```
model <- glm(Flag ~ X1 * X2, family = binomial, data = df_expanded)
summary(model)  #we can see the coefficients: α, β^X1, β^X2, β^X1X2
```

```
##
## Call:
```

```
## glm(formula = Flag ~ X1 * X2, family = binomial, data = df_expanded)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.52609    0.11062   4.756 1.98e-06 ***
## X1          -1.08571    0.15916  -6.821 9.02e-12 ***
## X2          -0.05609    0.21151  -0.265    0.791
## X1:X2        1.93746    0.29916   6.476 9.40e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1366.6  on 999  degrees of freedom
## Residual deviance: 1263.2  on 996  degrees of freedom
## AIC: 1271.2
##
## Number of Fisher Scoring iterations: 4
```

Now we check the results using the second method, computing log-odds.

```
# The first step is to create the theoretical distribution in RStudio.
table <- matrix(c(0.13, 0.21, 0.34, 0.05, 0.04, 0.09, 0.22, 0.12, 0.34, 0.08, 0.15
, 0.23, 0.35, 0.33, 0.68, 0.13, 0.19, 0.32), nrow = 3, ncol=6, byrow = TRUE)
rownames(table) <- c("Flag_0", "Flag_1", "Total")
colnames(table) <- c("MS_0N", "MS_1N","Total_N", "MS_0Y", "MS_1Y", "Total_Y")
table

##        MS_0N MS_1N Total_N MS_0Y MS_1Y Total_Y
## Flag_0  0.13  0.21    0.34  0.05  0.04    0.09
## Flag_1  0.22  0.12    0.34  0.08  0.15    0.23
## Total   0.35  0.33    0.68  0.13  0.19    0.32
```

Log-odds with Flag = 1.

```
p00 <- (table[2,1]/table[3,1])
p01 <- (table[2,4]/table[3,4])
p10 <- (table[2,2]/table[3,2])
p11 <- (table[2,5]/table[3,5])

log_odds00 <- log(p00/(1-p00)) # 0.5260931
log_odds01 <- log(p01/(1-p01)) # 0.4700036
log_odds10 <- log(p10/(1-p10)) # -0.5596158
log_odds11 <- log(p11/(1-p11)) # 1.321756
```

Now we are able to compute the coefficients α, βX1, βX2,βX1X2.

```
alpha <-log_odds00                                  # 0.5260931
beta_X1 <- log_odds10 -alpha                        # -1.085709
beta_X2 <- log_odds01 - alpha                       # -0.05608947
beta_X1X2 <- log_odds11 - alpha - beta_X1 - beta_X2 # 1.937461
```

The coefficient alpha represents the intercept, which corresponds to the baseline log-odds of the outcome when both predictors, X_1 and X_2, are equal to 0.

The coefficient beta_X1 represents the effect of X_1 when X_2 is held at 0. The negative value of β_1 indicates that X_1 has a negative effect on the log-odds in the absence of X_2.

The coefficient beta_X2 represents the effect of X_2 when X_1 is 0. It suggests a very small negative effect of X_2 when X_1 is absent.

The coefficient beta_X1X2 captures the interaction effect between X_1 and X_2. It measures how much additional change there is in the log-odds when both X_1 and X_2 are 1, beyond the sum of their individual effects. Its value indicates a strong positive interaction between the two variables.

b) draw a chart having on the Y axis the log odds of the Flag, and on the X axis the X1 in a way that the two subgroups of X2 are singled out. Discuss if the model is saturated.

```
library(ggplot2)
```

```
## Warning: il pacchetto 'ggplot2' è stato creato con R versione 4.3.3
```

```
library(RColorBrewer)
```

```
## Warning: il pacchetto 'RColorBrewer' è stato creato con R versione 4.3.1
```

```
library(scales)
```

```
## Warning: il pacchetto 'scales' è stato creato con R versione 4.3.3
```

After loading the libraries we create a new dataframe containing the combination of the two categorical variables (X1 and "Owner") and their associated log-odds.

```
df_log <- data.frame(
  X1 = factor(c(0,1,0,1),
              labels = c("Never Married", "Married")),
  Owner     = factor(c("N","N","Y","Y"),
                     labels = c("Not Owner","Owner")),
  log_odds  = c( 0.5260, -0.5596, 0.4700, 1.3218)
)
```

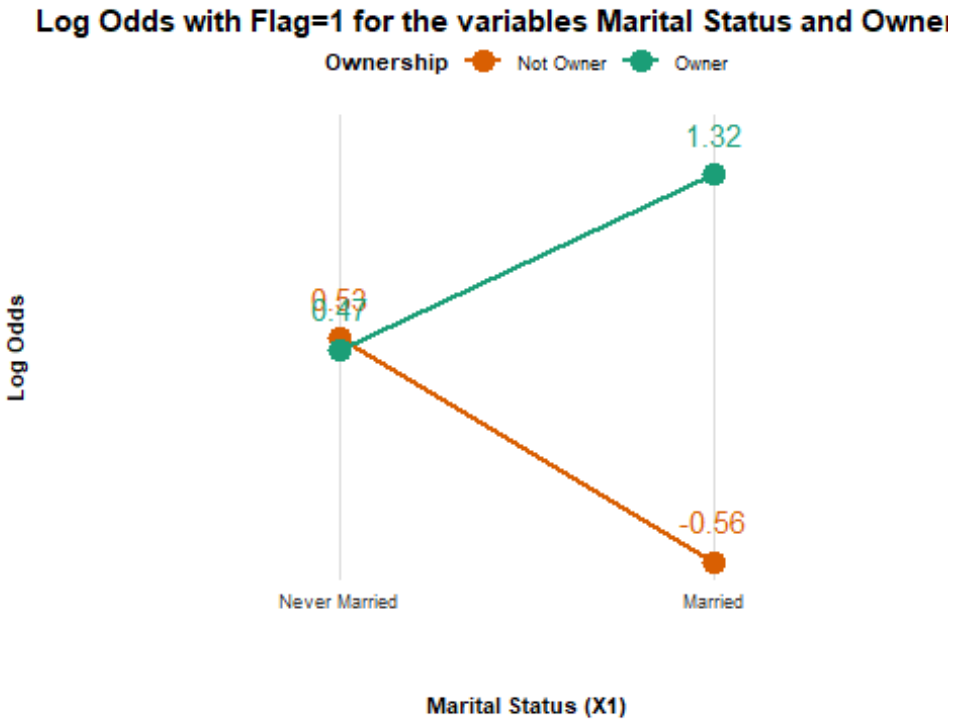
Now we create our plot.

```
#color palette
cols <- brewer.pal(n = 3, name = "Dark2")[2:1]

#plot
ggplot(df_log, aes(x = X1, y = log_odds, color = Owner, group = Owner)) +
  geom_line(size = 1, linetype = "solid") +
  geom_point(size = 3, shape = 21, stroke = 1.2, aes(fill = Owner)) +
  geom_text(aes(label = sprintf("%.2f", log_odds)),
            vjust = -1.2, size = 4, show.legend = FALSE) +
  scale_color_manual(values = cols) +
  scale_fill_manual(values = cols) +
  scale_y_continuous(
    breaks = sort(unique(df$log_odds)),
    labels = number_format(accuracy = 0.01),
```

```
    expand = expansion(mult = c(0.05, 0.15))
  ) +
  labs(
    title = "Log Odds with Flag=1 for the variables Marital Status and Ownership",
    x     = "Marital Status (X1)",
    y     = "Log Odds",
    color = "Ownership",
    fill  = "Ownership"
  ) +
  theme_minimal(base_size = 8) +
  theme(
    plot.title       = element_text(face = "bold", size = 11, hjust = 0.5, vjust =
-2.5),
    axis.title.x     = element_text(face = "bold", margin = margin(t = 40), vjust
= 7, size = 8),
    axis.title.y     = element_text(face = "bold", margin = margin(r = 40), vjust
= -3, size = 8),
    axis.text        = element_text(color = "#333333"),
    legend.position  = "top",
    legend.title     = element_text(face = "bold"),
    panel.grid.major = element_line(color = "#DDDDDD"),
    panel.grid.minor = element_blank()
  )
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



The plot shows the log-odds of the outcome (Flag = 1) based on marital status (X1) and home ownership (X2). The green line represents owners, and the orange line represents non-owners.

For owners, the log-odds increase from approximately 0.47 for “Never Married” to about 1.32 for “Married,” indicating that being married increases the likelihood of the outcome.

For non-owners, the log-odds decrease from approximately 0.53 for “Never Married” to about -0.56 for “Married,” indicating that being married decreases the likelihood of the outcome.

The lines are not parallel, suggesting an interaction effect between marital status and ownership on the log-odds, as confirmed by the previous point.

c) discuss what changes if instead of level ”N” of Owner you choose ”Y” as a reference level.

Now we impose our reference level as X2=1, instead of X2=0. We do the same computations as in point a) to analyze the changes in the model.

```
df_expanded$X2 <- factor(df_expanded$X2, levels = c(1, 0))  # ref = "Y"

#New logistic regression with the new reference level
model_newref <- glm(Flag ~ X1 * X2, family = binomial, data = df_expanded)
summary(model_newref)

##
## Call:
## glm(formula = Flag ~ X1 * X2, family = binomial, data = df_expanded)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.47000    0.18028   2.607 0.009131 **
```

```
## X1           0.85175    0.25331   3.362 0.000772 ***
## X20          0.05609    0.21151   0.265 0.790869
## X1:X20      -1.93746    0.29916  -6.476  9.4e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1366.6  on 999  degrees of freedom
## Residual deviance: 1263.2  on 996  degrees of freedom
## AIC: 1271.2
##
## Number of Fisher Scoring iterations: 4
```

By changing the reference level of X_2 from "N" (Not Owner) to "Y" (Owner), the interpretation of the coefficients must be updated accordingly. This shift alters the baseline category, requiring us to recalculate and reinterpret the log-odds for each combination of predictors within the new reference framework.

```
alpha <- log_odds01                                  # 0.4700036
beta_X1 <- log_odds11 - alpha                        # 0.8517522
beta_X2 <- log_odds00 - alpha                        # 0.05608947
beta_X1X2 <- log_odds10 - alpha - beta_X1 - beta_X2  # -1.937461
```

When changing the reference level of X_2 from "N" (Not Owner) to "Y" (Owner), the interpretation of the coefficients changes accordingly: + The intercept decreases slightly (from 0.5261 to 0.4700), reflecting the shift in baseline group. + β_1 changes from negative to positive (from −1.0857 to +0.8518), indicating a reversed effect of marital status depending on ownership. + β_2 flips sign (from −0.0561 to +0.0561) due to the new reference level. + β_{12} also flips sign (from +1.9375 to −1.9375), as expected with the reversed comparison.

d) discuss how the table should be modified in order for the term βX1 X2 to be equal zero.

In our logistic regression model, we included the interaction term between X_1 and X_2. This term captures any combined effect that goes beyond the sum of their individual contributions. For the interaction to be absent ($\beta_{x1x2} = 0$), the effect of one variable on the outcome must be consistent regardless of the value of the other. We can check it by using the cross-product ratio (CPR); if CPR = 1, the variables act independently with no interaction. Moreover, to achieve $\beta_{x1x2} = 0$, the table would need to be adjusted so that the effects of X_1 and X_2 are purely additive, meaning they influence the log-odds independently of one another.

First we calculate new joint conditional probabilities under the assumption that Flag and Marital Status are conditionally independent given Owner status (N or Y). It derives conditional probabilities of Flag and Marital Status within each Owner group by normalizing observed joint frequencies. Then, the joint conditional probabilities are estimated as the product of these conditionals, reflecting zero interaction between Flag and Marital Status within each Owner category. After this, we multiply these joint conditionals by the marginal probability of Owner produces updated joint probabilities across all variables. Finally, we can compute new log-odds for Flag=1 within each Owner and Marital Status combination, reflecting the adjusted association patterns after assuming conditional independence. By this way we model independence without losing the overall probability totals.

```r
#Marginal probabilities
marg_probNMS0 <- 0.35
marg_probNMS1 <- 0.33
marg_probNF0 <- 0.34
marg_probNF1 <- 0.34
marg_probN <- 0.68

marg_probYMS0 <- 0.13
marg_probYMS1 <- 0.19
marg_probYF0 <- 0.09
marg_probYF1 <- 0.23
marg_probY <- 0.32

#Conditional probabilities
#N
Pf0N <- (0.13+0.21)/marg_probN
Pf1N <- (0.22+0.12)/marg_probN
Pms0N <- (0.13+0.22)/marg_probN
Pms1N <- (0.21+0.12)/marg_probN

#Y
Pf0Y <- (0.05+0.04)/marg_probY
Pf1Y <- (0.08+0.15)/marg_probY
Pms0Y <- (0.05+0.08)/marg_probY
Pms1Y <- (0.04+0.15)/marg_probY

#New joint conditional probabilities
#N
CPN00 <- Pf0N * Pms0N
CPN01 <- Pf0N * Pms1N
CPN10 <- Pf1N * Pms0N
CPN11 <- Pf1N * Pms1N

#Y
CPY00 <- Pf0Y * Pms0Y
CPY01 <- Pf0Y * Pms1Y
CPY10 <- Pf1Y * Pms0Y
CPY11 <- Pf1Y * Pms1Y

#Joint probabilities
#N
N00 <-marg_probN * CPN00
N01 <-marg_probN * CPN01
N10 <-marg_probN * CPN10
N11 <-marg_probN * CPN11

#Y
Y00 <-marg_probY * CPY00
Y01 <-marg_probY * CPY01
```

```
Y10 <-marg_probY * CPY10
Y11 <-marg_probY * CPY11

#We create a new table using the new probabilities
tablenew <- matrix(c(N00,N01,marg_probNF0,Y00,Y01,marg_probYF0,N10,N11,marg_probNF
1,Y10,Y11,marg_probYF1,marg_probNMS0,marg_probNMS1,marg_probN,marg_probYMS0,marg_p
robYMS1,marg_probY), nrow = 3, ncol=6, byrow = TRUE)
rownames(tablenew) <- c("Flag_0", "Flag_1", "Total")
colnames(tablenew) <- c("MS_0N", "MS_1N","Total_N", "MS_0Y", "MS_1Y", "Total_Y")
tablenew

##        MS_0N MS_1N Total_N     MS_0Y     MS_1Y Total_Y
## Flag_0 0.175 0.165    0.34 0.0365625 0.0534375    0.09
## Flag_1 0.175 0.165    0.34 0.0934375 0.1365625    0.23
## Total  0.350 0.330    0.68 0.1300000 0.1900000    0.32

#New log-ODDS (with Flag=1)
p00new <- (tablenew[2,1]/tablenew[3,1])
p01new <- (tablenew[2,4]/tablenew[3,4])
p10new <- (tablenew[2,2]/tablenew[3,2])
p11new <- (tablenew[2,5]/tablenew[3,5])

log_odds00new <- log(p00new/(1-p00new))
log_odds01new <- log(p01new/(1-p01new))
log_odds10new <- log(p10new/(1-p10new))
log_odds11new <- log(p11new/(1-p11new))

# To check the indendence we use this theorem: if X1 and X2 are independent, then
p12(x1,x2)=p1(x1)p2(x2) for each x1 and x2.
#Therefore the cpr(X1,X2) has to be =1.
cpr_X1_X2 <- (p00new*p11new)/(p01new*p10new)
cpr_X1_X2

## [1] 1
```

Now we evaluate the coefficients to check if $\beta_{x1x2} = 0$.

```
alphanew <-log_odds00new
beta_X1new <- log_odds10new - alphanew
beta_X2new <- log_odds01new - alphanew
beta_X1X2new <- log_odds11new - alphanew - beta_X1new - beta_X2new #1.110223e-15≈0
```

EXERCISE 2

Input in R the dataset datiRETRO_TUSCANY_stud.txt. The dataset contains a retrospective sample of SMEs from Tuscany that were active in 2006, i.e. with positive revenues. The "SG" variable measures whether or not the firm was "Active" or "Bankrupted" at May 2010. Then, focusing only the following variables:

UN06; ICIP06; IIB06; IIF06; IIL06; IL06; OF06; RI06; RCI06; ID06; EBITDAV06;

```
#Download the dataset
dati<- read.table(choose.files(), header = TRUE, comment.char = "#")

#Import libraries
library(visdat)
```

```
## Warning: il pacchetto 'visdat' è stato creato con R versione 4.3.3
```

```
library(MASS)
library(Hmisc)
```

```
## Warning: il pacchetto 'Hmisc' è stato creato con R versione 4.3.3
```

```
##
## Caricamento pacchetto: 'Hmisc'
```

```
## I seguenti oggetti sono mascherati da 'package:base':
##
##     format.pval, units
```

```
library(latticeExtra)
```

```
## Warning: il pacchetto 'latticeExtra' è stato creato con R versione 4.3.3
```

```
## Caricamento del pacchetto richiesto: lattice
```

```
##
## Caricamento pacchetto: 'latticeExtra'
```

```
## Il seguente oggetto è mascherato da 'package:ggplot2':
##
##     layer
```

a) check if the flag is well constructed, by building a cross-table with SG. After a linearity check (details omitted), it can be seen that: ICIP06, EBITDAV06 and IL06 are better inputed in both linear and quadratic form. You then you may create (a) create ICI062 as the square of ICIP06 and add it to the list; alternatively (b) you may insert in the glm call of the square by adding I(name.variable)^2).

```
#cross-table with SG to check if the flag is well constructed
cross_table <- table(dati$flag, dati$SG)
cross_table
```

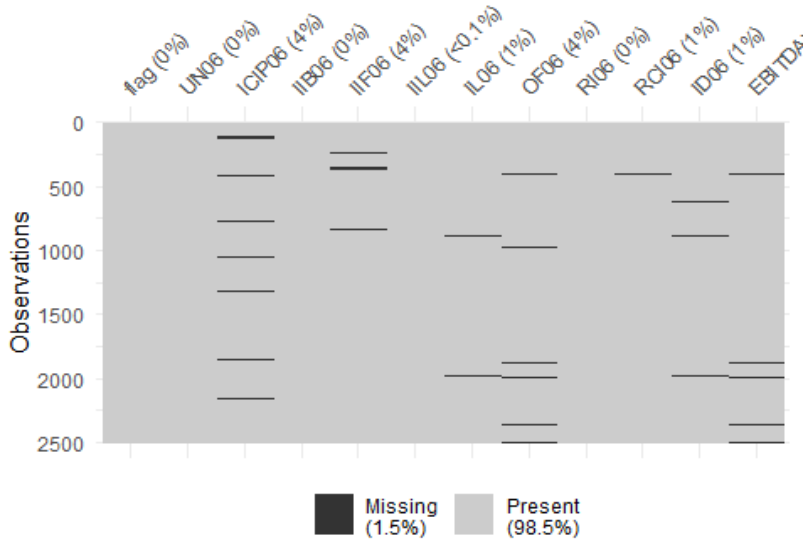
```
##
##     Ditta attiva Ditta in fallim
##   0            0             501
##   1         2004               0
```

The flag is well-constructed with respect to company status. When the flag is 0, the company is in "fallimento" (bankruptcy) in all 501 cases, and when the flag is 1, the company is "attiva" (active) in all 2004 cases.

```
#Important variables
var_imp <- c("flag","UN06","ICIP06","IIB06","IIF06","IIL06","IL06","OF06","RI06","
RCI06","ID06","EBITDAV06")
datin<-dati[,var_imp]
```

The new dataframe called *datin* is a subset of columns from the original dataframe (dati) that contains only the columns specified in the var_imp list.

```
vis_miss(datin)
```



This plot reveals that the dataset has some missing values, totaling 1.5% of the data. The horizontal black lines indicate the specific observations where these values are absent. While "flag" and "RI06" are completely present (0% missing), several other variables show gaps. Notably, "ICIP06", "IIF06", and "OF06" each have 4% missing data, followed by "RCI06" and "ID06" with 1% missing. "IIL06" has less than 0.1% missing data.

```
a <- complete.cases(datin)
table(a)

## a
## FALSE  TRUE
##   319  2186
```

The output from table(a) reveals that out of the total observations in datin, 2186 rows are complete (contain no missing values), while 319 rows have at least one missing value. This confirms the presence of missing data.

```
datin_c<-datin[a,]

b <- complete.cases(datin_c)
table(b)

## b
## TRUE
## 2186
```

The code creates datin_c by selecting only the rows from datin that are identified as complete (where a was TRUE). As a result, table(b) now confirms that datin_c consists of 2186 observations, all of which are complete and contain no missing values.

Estimate the first model (call it mod.iniz) that takes all variables above into account; evaluate if one single variable, choosen at your own interest, but in a way that the model remains hierarchical, can be omitted by looking at the p-value of the LR test, via the anova.

anova(mod1,mod2,test='Chisq')

in which mod1 is the reduced model obtained from mod2; compare the p-value of this test with the p-value of the Wald test.

```
mod.iniz <- glm(flag ~ UN06 + ICIP06 + I(ICIP06^2) + IIB06 + IIF06 + IIL06 + IL06
+ I(IL06^2) + OF06 + RCI06 + RI06 + ID06 + EBITDAV06 + I(EBITDAV06^2), data = dati
n_c, family = binomial)

mod.reduced <- glm(flag ~ UN06 + IIB06 + IIF06 + IIL06 + IL06 + I(IL06^2) + OF06 +
RCI06 + RI06 + ID06 + EBITDAV06 + I(EBITDAV06^2), data = datin_c, family = binomia
l)

#ANOVA TEST
anova(mod.reduced,mod.iniz, test='Chisq')

## Analysis of Deviance Table
##
## Model 1: flag ~ UN06 + IIB06 + IIF06 + IIL06 + IL06 + I(IL06^2) + OF06 +
##     RCI06 + RI06 + ID06 + EBITDAV06 + I(EBITDAV06^2)
## Model 2: flag ~ UN06 + ICIP06 + I(ICIP06^2) + IIB06 + IIF06 + IIL06 +
##     IL06 + I(IL06^2) + OF06 + RCI06 + RI06 + ID06 + EBITDAV06 +
##     I(EBITDAV06^2)
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      2174     1608.1
## 2      2172     1608.1  2 0.075274   0.9631
```

The ANOVA test comparing mod.reduced (Model 1) and mod.iniz (Model 2) confirms that removing ICIP06 and I(ICIP06^2) from the model does not significantly impact its fit. The large p-value of 0.9631 (Pr(>Chi)) indicates that there's no statistically significant difference between the two

models. This means the additional terms in mod.iniz (the ICIP06 variables) do not add significant explanatory power to the model, supporting the decision to use the simpler mod.reduced.

```
# Wald test p-value for ICIP06 in the full model
summary(mod.iniz)$coefficients["ICIP06", "Pr(>|z|)"]

## [1] 0.8699752
```

The output 0.8699752 is the p-value from the Wald test for the ICIP06 coefficient in the mod.iniz. A p-value this high indicates that ICIP06 is not statistically significant in predicting the flag variable when other predictors are also in the model. This result reinforces the conclusion from the ANOVA test that removing ICIP06 and its squared term had no significant impact on the model's fit.

c) After selecting only the complete cases, choose the best model via the backward procedure.

mod.fin<-stepAIC(mod.iniz,direction=''backward'')

notice that this procedure uses by default the AIC criterion. Comment the final model, also by looking at the sign of the variables, and seeing if it goes in the expected direction.

```
mod.fin <- stepAIC(mod.iniz, direction = "backward", test = "Chisq")

## Start:  AIC=1636.07
## flag ~ UN06 + ICIP06 + I(ICIP06^2) + IIB06 + IIF06 + IIL06 +
##     IL06 + I(IL06^2) + OF06 + RCI06 + RI06 + ID06 + EBITDAV06 +
##     I(EBITDAV06^2)
##
##
## Step:  AIC=1636.07
## flag ~ UN06 + ICIP06 + I(ICIP06^2) + IIB06 + IIF06 + IL06 + I(IL06^2) +
##     OF06 + RCI06 + RI06 + ID06 + EBITDAV06 + I(EBITDAV06^2)
##
##                  Df Deviance    AIC     LRT   Pr(Chi)
## - ICIP06          1   1608.1 1634.1   0.027 0.8699235
## - I(ICIP06^2)     1   1608.1 1634.1   0.071 0.7899252
## - OF06            1   1608.2 1634.2   0.095 0.7581023
## - ID06            1   1608.2 1634.2   0.135 0.7134104
## - UN06            1   1608.8 1634.8   0.682 0.4088554
## - I(IL06^2)       1   1609.2 1635.2   1.101 0.2941479
## - IL06            1   1609.3 1635.3   1.283 0.2574090
## - RI06            1   1609.4 1635.4   1.372 0.2414840
## <none>                1608.1 1636.1
## - I(EBITDAV06^2)  1   1612.1 1638.1   4.044 0.0443345 *
## - EBITDAV06       1   1622.1 1648.1  14.041 0.0001789 ***
## - IIB06           1   1627.6 1653.6  19.535 9.879e-06 ***
## - RCI06           1   1645.4 1671.4  37.338 9.931e-10 ***
## - IIF06           1   1759.6 1785.6 151.492 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1634.1
## flag ~ UN06 + I(ICIP06^2) + IIB06 + IIF06 + IL06 + I(IL06^2) +
##     OF06 + RCI06 + RI06 + ID06 + EBITDAV06 + I(EBITDAV06^2)
```

```
##
##                   Df Deviance    AIC     LRT   Pr(Chi)
## - I(ICIP06^2)      1   1608.1 1632.1   0.048 0.8257676
## - OF06             1   1608.2 1632.2   0.101 0.7503722
## - ID06             1   1608.2 1632.2   0.123 0.7262623
## - UN06             1   1608.8 1632.8   0.687 0.4070922
## - I(IL06^2)        1   1609.2 1633.2   1.124 0.2890427
## - IL06             1   1609.4 1633.4   1.296 0.2548864
## - RI06             1   1609.7 1633.7   1.579 0.2089655
## <none>                 1608.1 1634.1
## - I(EBITDAV06^2)   1   1612.1 1636.1   4.022 0.0449107 *
## - EBITDAV06        1   1622.1 1646.1  14.038 0.0001792 ***
## - IIB06            1   1627.8 1651.8  19.736 8.893e-06 ***
## - RCI06            1   1645.5 1669.5  37.356 9.842e-10 ***
## - IIF06            1   1768.5 1792.5 160.363 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1632.14
## flag ~ UN06 + IIB06 + IIF06 + IL06 + I(IL06^2) + OF06 + RCI06 +
##     RI06 + ID06 + EBITDAV06 + I(EBITDAV06^2)
##
##                   Df Deviance    AIC     LRT   Pr(Chi)
## - OF06             1   1608.2 1630.2   0.095 0.7573587
## - ID06             1   1608.3 1630.3   0.171 0.6788117
## - UN06             1   1608.8 1630.8   0.690 0.4062657
## - I(IL06^2)        1   1609.2 1631.2   1.082 0.2981503
## - IL06             1   1609.4 1631.4   1.267 0.2602479
## - RI06             1   1609.7 1631.7   1.530 0.2160713
## <none>                 1608.1 1632.1
## - I(EBITDAV06^2)   1   1612.2 1634.2   4.046 0.0442756 *
## - EBITDAV06        1   1622.2 1644.2  14.056 0.0001774 ***
## - IIB06            1   1629.3 1651.3  21.119 4.316e-06 ***
## - RCI06            1   1645.5 1667.5  37.315 1.005e-09 ***
## - IIF06            1   1768.5 1790.5 160.368 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1630.24
## flag ~ UN06 + IIB06 + IIF06 + IL06 + I(IL06^2) + RCI06 + RI06 +
##     ID06 + EBITDAV06 + I(EBITDAV06^2)
##
##                   Df Deviance    AIC     LRT   Pr(Chi)
## - ID06             1   1608.4 1628.4   0.168 0.6822838
## - UN06             1   1608.9 1628.9   0.664 0.4153108
## - I(IL06^2)        1   1609.3 1629.3   1.054 0.3046343
## - IL06             1   1609.4 1629.4   1.199 0.2734464
## - RI06             1   1609.8 1629.8   1.524 0.2170778
## <none>                 1608.2 1630.2
## - I(EBITDAV06^2)   1   1612.8 1632.8   4.566 0.0326071 *
```

```
## - EBITDAV06       1   1623.3 1643.3  15.020 0.0001064 ***
## - IIB06           1   1631.2 1651.2  22.988 1.630e-06 ***
## - RCI06           1   1647.4 1667.4  39.164 3.898e-10 ***
## - IIF06           1   1768.7 1788.7 160.466 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1628.41
## flag ~ UN06 + IIB06 + IIF06 + IL06 + I(IL06^2) + RCI06 + RI06 +
##     EBITDAV06 + I(EBITDAV06^2)
##
##                  Df Deviance    AIC     LRT   Pr(Chi)
## - UN06            1   1609.0 1627.0   0.638 0.4244949
## - IL06            1   1609.4 1627.4   1.036 0.3087816
## - I(IL06^2)       1   1609.7 1627.7   1.244 0.2647284
## - RI06            1   1610.1 1628.1   1.684 0.1943420
## <none>                1608.4 1628.4
## - I(EBITDAV06^2)  1   1613.0 1631.0   4.548 0.0329643 *
## - EBITDAV06       1   1623.5 1641.5  15.059 0.0001042 ***
## - IIB06           1   1634.0 1652.0  25.596 4.208e-07 ***
## - RCI06           1   1647.6 1665.6  39.199 3.827e-10 ***
## - IIF06           1   1778.9 1796.9 170.474 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1627.05
## flag ~ IIB06 + IIF06 + IL06 + I(IL06^2) + RCI06 + RI06 + EBITDAV06 +
##     I(EBITDAV06^2)
##
##                  Df Deviance    AIC     LRT   Pr(Chi)
## - IL06            1   1610.1 1626.1   1.090   0.29658
## - I(IL06^2)       1   1610.3 1626.3   1.273   0.25912
## - RI06            1   1610.7 1626.7   1.658   0.19794
## <none>                1609.0 1627.0
## - I(EBITDAV06^2)  1   1613.8 1629.8   4.729   0.02966 *
## - EBITDAV06       1   1625.2 1641.2  16.143 5.873e-05 ***
## - IIB06           1   1634.7 1650.7  25.609 4.181e-07 ***
## - RCI06           1   1648.6 1664.6  39.513 3.258e-10 ***
## - IIF06           1   1782.6 1798.6 173.513 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1626.14
## flag ~ IIB06 + IIF06 + I(IL06^2) + RCI06 + RI06 + EBITDAV06 +
##     I(EBITDAV06^2)
##
##                  Df Deviance    AIC     LRT   Pr(Chi)
## - I(IL06^2)       1   1610.3 1624.3   0.209   0.64737
## - RI06            1   1611.8 1625.8   1.684   0.19439
## <none>                1610.1 1626.1
```

```
## - I(EBITDAV06^2)  1   1614.6 1628.6   4.440   0.03511 *
## - EBITDAV06       1   1626.0 1640.0  15.823 6.957e-05 ***
## - IIB06           1   1636.9 1650.9  26.799 2.257e-07 ***
## - RCI06           1   1655.9 1669.9  45.780 1.323e-11 ***
## - IIF06           1   1811.9 1825.9 201.736 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1624.34
## flag ~ IIB06 + IIF06 + RCI06 + RI06 + EBITDAV06 + I(EBITDAV06^2)
##
##                  Df Deviance    AIC     LRT   Pr(Chi)
## - RI06            1   1612.0 1624.0   1.655   0.19824
## <none>                1610.3 1624.3
## - I(EBITDAV06^2)  1   1614.9 1626.9   4.536   0.03319 *
## - EBITDAV06       1   1626.5 1638.5  16.148 5.859e-05 ***
## - IIB06           1   1637.7 1649.7  27.375 1.676e-07 ***
## - RCI06           1   1655.9 1667.9  45.577 1.468e-11 ***
## - IIF06           1   1822.3 1834.3 211.935 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=1624
## flag ~ IIB06 + IIF06 + RCI06 + EBITDAV06 + I(EBITDAV06^2)
##
##                  Df Deviance    AIC     LRT   Pr(Chi)
## <none>                1612.0 1624.0
## - I(EBITDAV06^2)  1   1616.4 1626.4   4.362   0.03674 *
## - EBITDAV06       1   1627.5 1637.5  15.544 8.063e-05 ***
## - IIB06           1   1639.7 1649.7  27.698 1.418e-07 ***
## - RCI06           1   1659.0 1669.0  47.040 6.956e-12 ***
## - IIF06           1   1836.8 1846.8 224.791 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

stepAIC() refined the initial model by progressively removing non-contributing predictors, leading to a model with a lower AIC, indicating a better balance between fit and complexity. The resulting model focuses on variables like IIB06, IIF06, RCI06, and EBITDAV06 (and its squared term) as the key predictors for flag.

d) Test the assumption H0: "the true model is the one that contains the intercept only" against H1: "the true model is the one selected by the backward procedure". Do it first by separately estimating the two models and using the anova command.

```r
#Null model
mod.null <- glm(flag ~ 1, family = binomial, data = datin_c)

#Likelihood-ratio test (chi-squared)
anova(mod.null, mod.fin, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: flag ~ 1
## Model 2: flag ~ IIB06 + IIF06 + RCI06 + EBITDAV06 + I(EBITDAV06^2)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      2185     1981.6
## 2      2180     1612.0  5   369.61 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The very low p-value from the ANOVA test demonstrates that mod.fin provides a significantly better fit to the data than the null model (intercept only). This confirms that the chosen predictors (IIB06, IIF06, RCI06, EBITDAV06, and I(EBITDAV06^2)) are indeed useful in explaining the variation in the flag variable.

Show that Null Deviance of mod.fin is exactly the value of the Deviance of the anova test of current model against the model with the intercept only (that means that in the future you do not need to estimate such a model!); comment on the result of this test and explain the values of the degrees of freedom.

The ANOVA test confirms that our final model (mod.fin) is a significantly better fit than the simpler, intercept-only model. Crucially, the Null Deviance reported for mod.fin is precisely the same as the residual deviance of the null model in the ANOVA table. This means you inherently have the baseline deviance for comparison without needing to estimate a separate null model. The extremely low p-value (<2.2e−16) clearly indicates that the predictors added to mod.fin substantially improve the model's ability to explain the 'Flag' variable. The 5 degrees of freedom in the test simply reflect the five additional predictor variables incorporated into mod.fin.

e) Use the umbria_match_stud.txt as a validation sample. Calculate the estimated score function using the model selected at c). Do it wither using the matrix multiplication or using the predict. Notice that in the first case, if you in your score function you have a variable inserted in the quadratic form not inserted in the call, you have to expand the validation dataset by adding the relevant column (in the correct position).

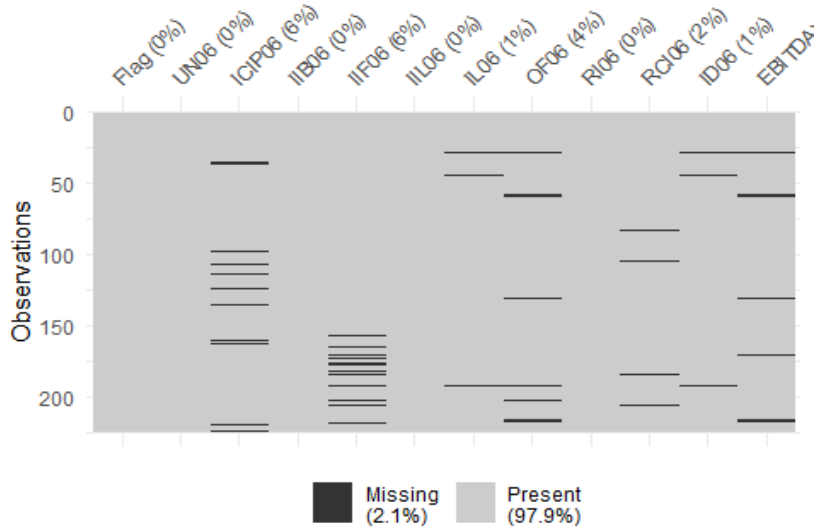
```
#Download the dataset
dati_val <- read.csv("~/umbria_match_stud (4).csv", sep=";")

#Computation of the Flag
dati_val$Flag<-dati_val$SG=="Ditta attiva"
table(dati_val$Flag,dati_val$SG)

##
##         Ditta attiva Ditta in fallim
##   FALSE            0              56
##   TRUE           168               0

#Important variables
var_imp_val <- c("Flag","UN06","ICIP06","IIB06","IIF06","IIL06","IL06","OF06","RI0
6","RCI06","ID06","EBITDAV06")
datin_val <-dati_val[,var_imp_val]
```

```
vis_miss(datin_val)
```



The *vis_miss()* plot for the validation dataset reveals the presence of missing values across several variables, totaling 2.1% of the data. While 'Flag', 'UN06', 'IIB06', and 'RI06' are complete, significant gaps exist in others. Specifically, 'ICIP06' and 'IIF06' each exhibit 6% missingness, 'OF06' has 4%, 'RCI06' has 2%, and 'IL06' and 'ID06' each have 1% missing data.

```
a_val <- complete.cases(datin_val)
table(a_val)

## a_val
## FALSE  TRUE
##    39   185

datin_c_val<-datin_val[a_val,]

b_val <- complete.cases(datin_c_val)
table(b_val)

## b_val
## TRUE
##  185
```

The complete.cases(datin_val) command identified that 185 out of 224 observations in datin_val were complete, with 39 rows containing at least one missing value. Subsequently, datin_c_val <- datin_val[a_val,] was executed, creating a new dataframe, datin_c_val, which exclusively comprises these 185 complete observations.

```
#Calculate the estimated score function using the model selected at c
score_pred <- predict(mod.fin, newdata = datin_c_val, type = "response")

# Separate the predicted scores by the 'Flag' variable
```

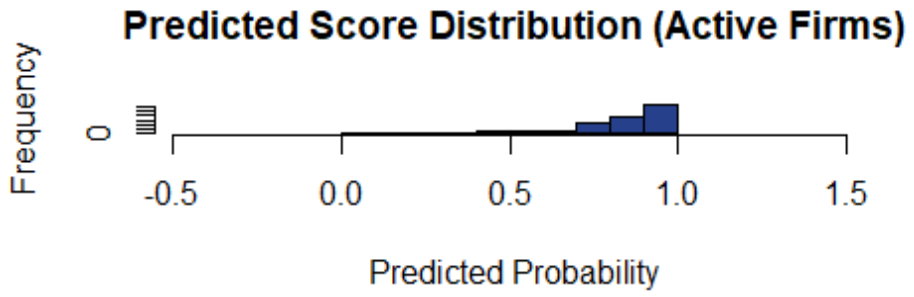
```r
score_active <- score_pred[datin_c_val$Flag == TRUE]
score_non_active <- score_pred[datin_c_val$Flag == FALSE]
```

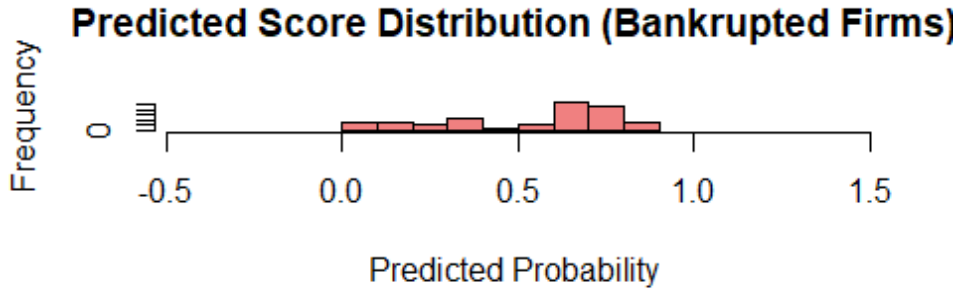
Using the same graphical tools of the Exercise 1 in the past Esercitazione 1, explore graphically if the score so built is good in discriminating between the two populations.

```r
# Histograms
par(mfrow = c(2, 1))

hist(score_active, col = "royalblue4", main = "Predicted Score Distribution (Activ
e Firms)", xlab = "Predicted Probability", ylab = "Frequency",
     xlim=range(min(score_active)-0.5, max(score_active)+0.5))

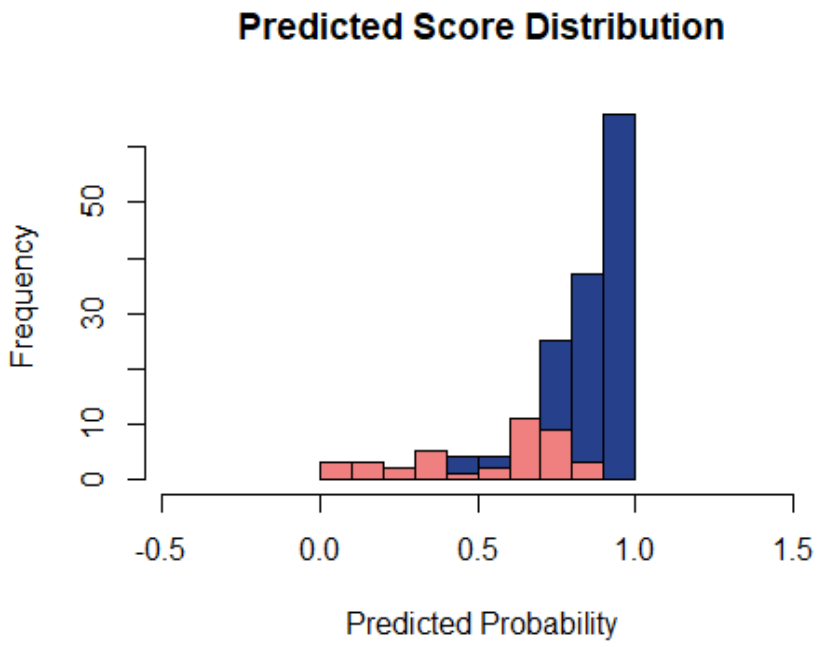
hist(score_non_active, col = "lightcoral", main = "Predicted Score Distribution (B
ankrupted Firms)", xlab = "Predicted Probability", ylab = "Frequency",
     xlim=range(min(score_non_active)-0.5, max(score_non_active)+0.54))
```



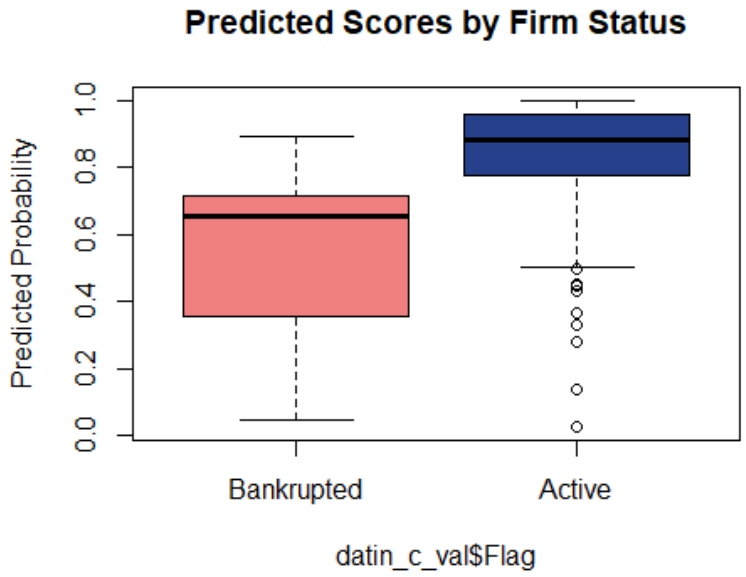



```r
par(mfrow = c(1, 1))

hist(score_active, col = "royalblue4", main = "Predicted Score Distribution", xlab
= "Predicted Probability", ylab = "Frequency",
     xlim=range(min(score_active)-0.5, max(score_active)+0.5))
hist(score_non_active, col = "lightcoral", main = "Predicted Score Distribution",
xlab = "Predicted Probability", ylab = "Frequency", add = TRUE,
     xlim=range(min(score_non_active)-0.5, max(score_non_active)+0.5))
```

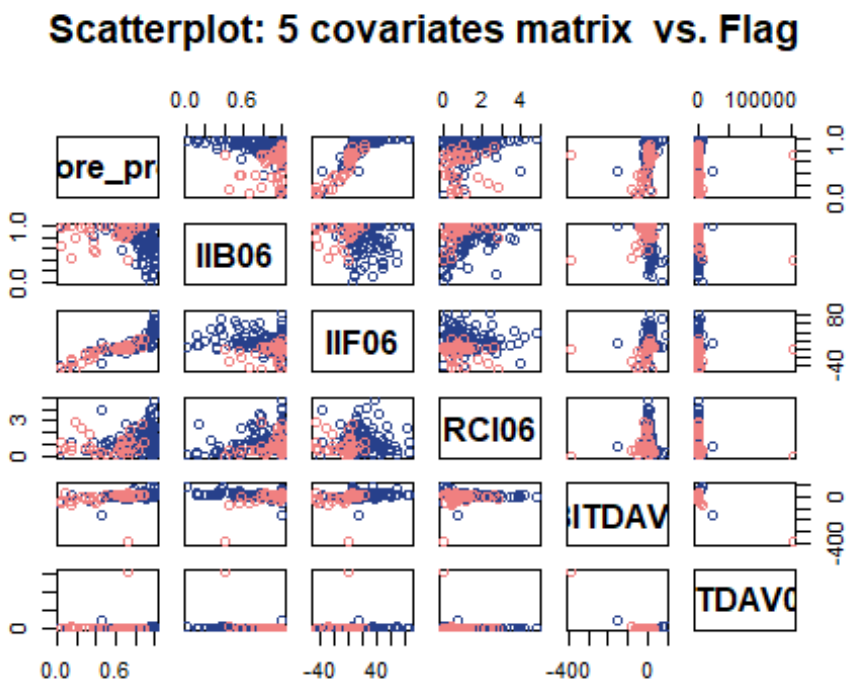



```
# Box plot
boxplot(score_pred ~ datin_c_val$Flag, col = c("lightcoral", "royalblue4"), names
= c("Bankrupted", "Active"), ylab = "Predicted Probability", main = "Predicted Sco
res by Firm Status")
```




```
#Scatterplot
pairs(score_pred ~ IIB06 + IIF06 + RCI06 + EBITDAV06 + I(EBITDAV06^2),
      data = datin_c_val,
      col = ifelse(datin_c_val$Flag == TRUE, "royalblue4", "lightcoral"),
```

```
      main = "Scatterplot: 5 covariates matrix  vs. Flag",
      cex.labels = 1.5,
      font.labels = 2)
```



The model shows some separation, with active firms generally predicted at higher probabilities and bankrupted firms at lower ones. However, the significant overlap in the mid-range of predicted scores indicates suboptimal discrimination. This suggests a notable portion of firms are likely to be misclassified, highlighting an area for model improvement.

f) Again in the validation sample, build the ROC curve and calculate the I_{roc} index.

```
library(pROC)
```

```
## Warning: il pacchetto 'pROC' è stato creato con R versione 4.3.3
```

```
## Type 'citation("pROC")' for a citation.
```

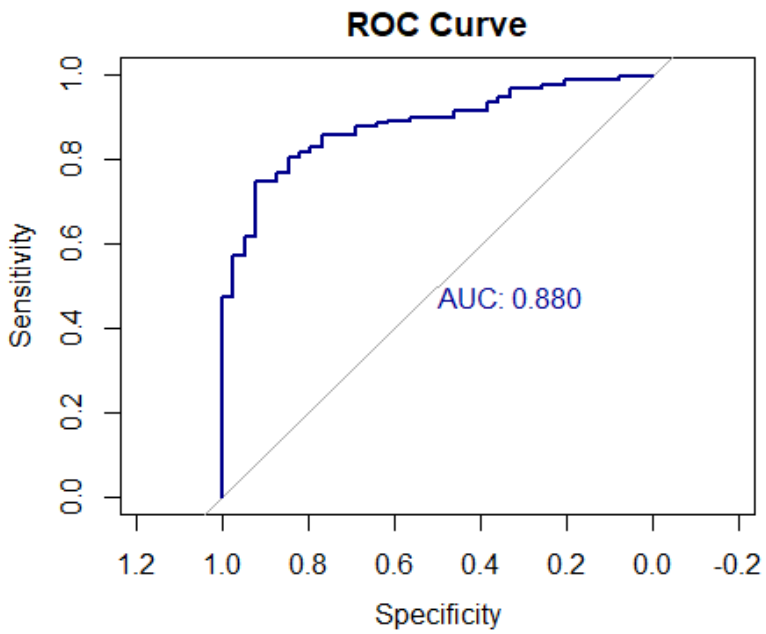
```
##
## Caricamento pacchetto: 'pROC'
```

```
## I seguenti oggetti sono mascherati da 'package:stats':
##
##     cov, smooth, var
```

```
roc <- roc(datin_c_val$Flag, score_pred, plot=TRUE, main="ROC Curve",
           col="blue4", print.auc = TRUE)
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```



The ROC curve visualizes the model's ability to distinguish between active and bankrupted firms across all possible thresholds. The AUC (Area Under the Curve) value of 0.880 indicates that the model has good discriminative power.

g) Evaluate the probability of non-default of an average firm (i.e. the firm that has the mean value for each significant indicator), after noting that the training sample is drawn in a retrospective way from a population with 345302 active firms and 10697 defaulting firms.

By Bayes' Theorem:

$$g(\theta \mid \mathbf{y}) = \frac{f(\mathbf{y} \mid \theta)p(\theta)}{f(\mathbf{y})}$$

Where the numerator is the product of the likelihood and the prior and the denominator is the marginal distribution of the data, we can identify y as the characteristics of the average firm and theta as the default/non-default state. In our case, we first define the prior probability using the population data. Then, we use the output from our previously created model as the likelihood (as f(y|θ)). Specifically, through the predict command, we obtain the conditional probability of non-default given the average firm's characteristics. Subsequently, we apply Bayes' Theorem to get the posterior probability that the average firm is non-defaulting, considering both the prior probability and the evidence provided by the mod.fin model. In our specific case, the formula is:

P(Non-Default | Average Firm)=(P(Average Firm | Non-Default)×P(Non-Default)) / P(Average Firm) where *P(Average Firm)* is given by
(P(Average Firm | Non-Default)×P(Non-Default)) + (P(Average Firm | Default)×P(Default))

```
# Calculation of the significant variables of the average company
average_firm_data <- data.frame(
  IIB06 = mean(datin_c$IIB06, na.rm = TRUE),
  IIF06 = mean(datin_c$IIF06, na.rm = TRUE),
  RCI06 = mean(datin_c$RCI06, na.rm = TRUE),
  EBITDAV06 = mean(datin_c$EBITDAV06, na.rm = TRUE),
```

```
  `I(EBITDAV06^2)` = mean(datin_c$EBITDAV06, na.rm = TRUE)^2
)
average_firm_data

##       IIB06    IIF06    RCI06 EBITDAV06 I.EBITDAV06.2.
## 1 0.8428271 18.88338 1.234296  8.489918        72.0787
```

In this part of the code, we identify an average company based on the averages of the significant indicators present in the dataset 'datin_c'.

```
#P(Average Firm|Non-Default)
prob_nondefault_averagefirm <- predict(mod.fin, newdata = average_firm_data, type
= "response")
prob_nondefault_averagefirm

##       1
## 0.89397

#P(Average Firm|Default)
prob_default_averagefirm <- 1 - prob_nondefault_averagefirm
prob_default_averagefirm

##       1
## 0.10603

#Prior probabilities P(ϑ)
total_firms_pop <- 345302 + 10697
defaulting_firms_pop <- 10697
non_defaulting_firms_pop <- 345302

prior_prob_default <- defaulting_firms_pop/ total_firms_pop
prior_prob_non_default <- non_defaulting_firms_pop/ total_firms_pop

#We calculate the denominator, which is the marginal probability of observing the
average firm's characteristics in the population
marginal_prob_averagefirm <- (prob_nondefault_averagefirm * prior_prob_non_default
) +
  (prob_default_averagefirm * prior_prob_default)

#Posterior probability that the average firm is not in default.
posterior_prob_non_default_average_firm <- (prob_nondefault_averagefirm * prior_pr
ob_non_default) / marginal_prob_averagefirm

prior_prob_non_default # Prior probability of non-default (population)

## [1] 0.9699522

prob_nondefault_averagefirm # Model's estimated probability of non-default (likeli
hood)

##       1
## 0.89397
```

```
posterior_prob_non_default_average_firm # Posterior probability of non-default for the average firm
```

```
##         1
## 0.9963392
```

The posterior probability result is very high. This could be due to the likelihood, as the ratio between the probability of observing the average firm's characteristics in a non-defaulting firm (P(average firm|Non-Default)) and the probability of observing the average firm's characteristics in a defaulting firm (P(average firm|Default)) is quite high (0.894/0.106=8.43). This means that the average firm's characteristics are 8.43 times more probable if the firm is non-defaulting compared to when it is defaulting. This evidence further strengthens a prior that was already high, leading to a very high posterior value.

EXERCISE3

a) perform a linearity check on the continuous variables.

```
#Import dataset
dati <- read.table(file.choose(), header=TRUE) #cartolarization data

#The continuous variables are num, Flag, Age, lastsal, lastral, avgsal, recvsaff
#kept, rpcrate, prorate
numeric<-sapply(dati, is.numeric)
numeric
```

```
##      num     Flag      Age     zres  lastsal  lastrat   avgsal recvsaff
##     TRUE     TRUE     TRUE    FALSE     TRUE     TRUE     TRUE     TRUE
##     kept     prot      imm  rpcrate  prorate
##     TRUE    FALSE    FALSE     TRUE     TRUE
```

Extraction of the continuous variable (Age) and the response variable (Flag) in order to be able to realize a linearity check on these two variables. After this, it will be necessary to repeat this procedure for all the continuous variables.

```
#AGE
x<-dati$Age
Flag<-factor(dati$Flag)
```

Computation of quantiles of "x", ignoring missing values (NA). We obtain the minimum, Q1, median, Q3 and the maximum.

```
qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)
```

We now create a variable, VAR_Q, with five levels: it groups each value of x (Age) into four quartile-based categories, assigning missing values to a fifth level, in order to segment Age and explore potential non-linear effects on Flag.

```
VAR_Q <- matrix(nrow = length(x), ncol=1)
VAR_Q[is.na(x)==TRUE]=5
#the values of x below the second quantile (0,25) have level 1
VAR_Q[x<=qnt[2]]<-1

#the values of x between the second quantile and the third one have level 2
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2

#the values of x between the third quantile and the fourth one have level 3
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3

#the values of x between the fourth quantile and the fifth one have level 4
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4

#we transform VAR_Q in factor
VARQ=factor(VAR_Q)
```

Since we have VARQ as a factor, we are able to build a logistic regression model and to extract the p-value of the coefficients. The evaluation of the p-value is important to assess the statistical significance of the differences between the levels of Age.

```
modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]
p_value
```

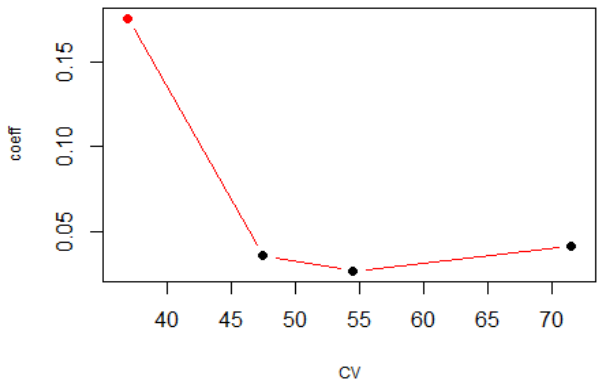
```
## (Intercept)       VARQ2       VARQ3       VARQ4
## 0.001056839 0.638682478 0.738321119 0.591838692
```

It's helpful to create a plot for a more intuitive visualization of the results, and a first step involves calculating the centers of the quantiles to add them to the x-axis of the graph.

```
#Assignment of colors for the plot
color=p_value
color[p_value>0.05]="black" #not significant coefficients
color[p_value<=0.05]="red"  #significant coefficients

#We compute the average values of the quantiles to place coefficients on the graph
CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

#plot
plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



The plot helps to check for linearity between Age and Flag: in a linear relationship, the coefficients should change in a consistent pattern. Here, however, only the first quartile appears to be significantly different, suggesting non-linearity.

```
#LASTSAL
x<-dati$lastsal
```

```
Flag<-factor(dati$Flag)

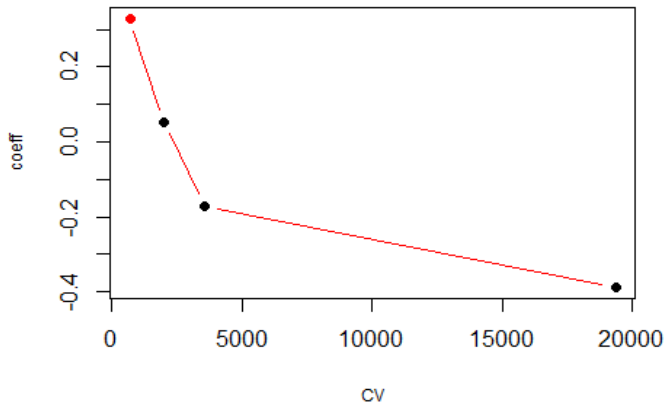
qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]

CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



The linearity check indicates a strong non-linear trend: the effect is positive in the lowest quartile, turns negative in the second, and becomes increasingly negative in the higher quartiles.

```
x<-dati$lastrat
Flag<-factor(dati$Flag)

qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

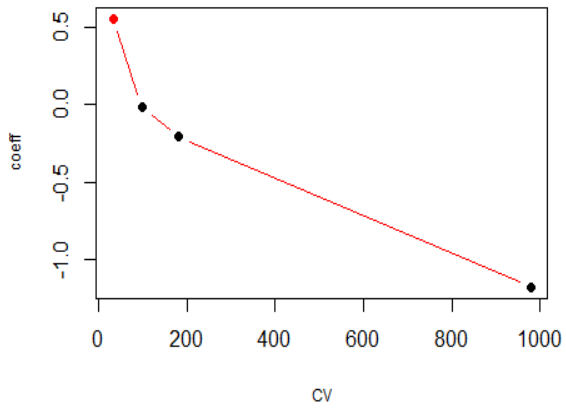
VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
```

```
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]

CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



The plot shows a clear decreasing, non-linear relationship.

```
#AVGSAL
x<-dati$avgsal
Flag<-factor(dati$Flag)

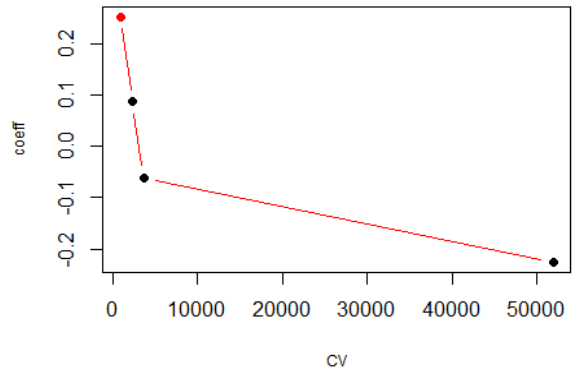
qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]
```

```
CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



This graph shows a linearity check that represents a non linear relationship between the variables.

```
#RECVSAFF
x<-dati$recvsaff
Flag<-factor(dati$Flag)

qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

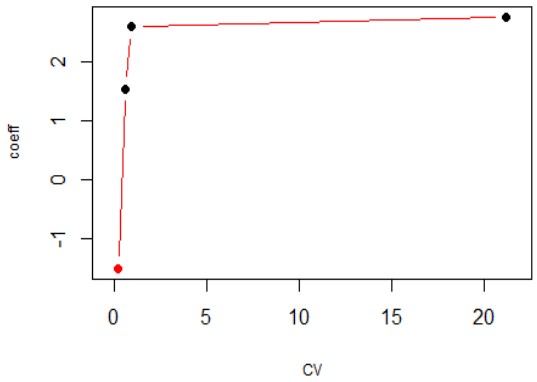
VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]

CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
```

```
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



The plot shows a clear increasing, non-linear relationship.

```
#KEPT
x<-dati$kept
Flag<-factor(dati$Flag)

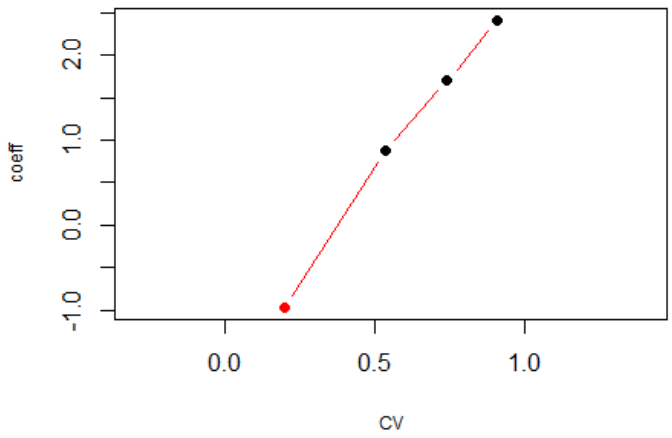
qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]

CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



The linearity check represents a linear relationship between “kept” and “Flag.”

```
#RPCRATE
x<-dati$rpcrate
Flag<-factor(dati$Flag)

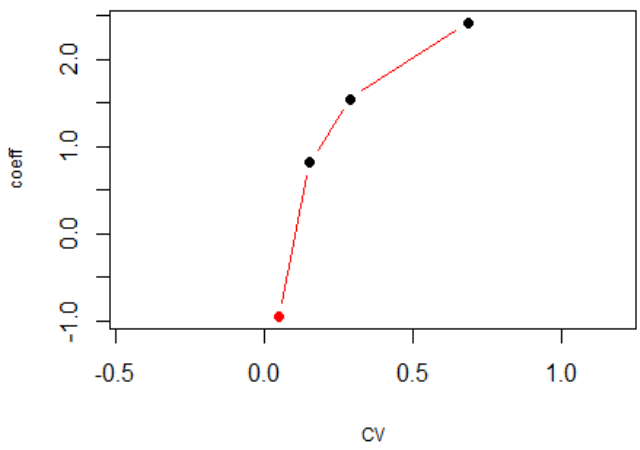
qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]

CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



The linearity check represents a positive non-perfectly linear relationship.

```
#PRORATE
x<-dati$prorate
Flag<-factor(dati$Flag)

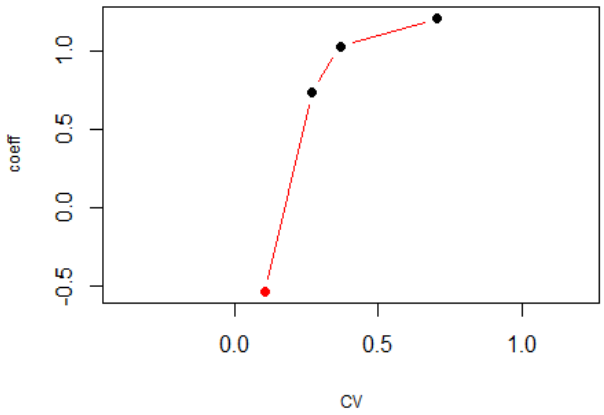
qnt = quantile(x, probs=c(0, 0.25, 0.50, 0.75, 1), na.rm = TRUE)

VAR_Q=matrix(nrow = length(x), ncol = 1)
VAR_Q[is.na(x)==TRUE]=5
VAR_Q[x<=qnt[2]]<-1
VAR_Q[x>qnt[2]&x<=qnt[3]]<-2
VAR_Q[x>qnt[3]&x<=qnt[4]]<-3
VAR_Q[x>qnt[4]&x<=qnt[5]]<-4
VARQ=factor(VAR_Q)

modl=glm(Flag~VARQ, family=binomial)
p_value=summary(modl)$coefficients[,4]

CV1=(qnt[1]+qnt[2])/2
CV2=(qnt[2]+qnt[3])/2
CV3=(qnt[3]+qnt[4])/2
CV4=(qnt[4]+qnt[5])/2
CV=c(CV1,CV2,CV3,CV4)

plot(CV, modl$coefficients[1:4], type = "b", pch = 19, lty = 1, col =color, ylab="
coeff",cex.lab=0.75, xlim=range(CV1-0.5,CV4+0.5), ylim=range(min(modl$coefficients
), max(modl$coefficients)))
```



There is a positive non linear relationship.

b) on a random basis, split the whole sample into a training sample of size 4000 and a validation sample of size 1276. On the training sample, conduct a preliminary assessment of the significance of variables by fitting separate univariate regression models.

```
#We prepare a dataset
str(dati)

## 'data.frame':    5276 obs. of  13 variables:
##  $ num     : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Flag    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ Age     : int  58 63 74 65 54 53 49 75 41 61 ...
##  $ zres    : chr  "NORD" "CENTRO" "NORD" "NORD" ...
##  $ lastsal : num  1315 5338 2203 175 3460 ...
##  $ lastrat : num  227 911 378 84 814 ...
##  $ avgsal  : num  1099 5209 2147 140 3674 ...
##  $ recvsaff: num  0.292 0 0.612 0 0.242 0.482 0.69 0 0.305 0 ...
##  $ kept    : num  0.529 0 0.529 0 0.667 0.645 0.5 0 0.667 0 ...
##  $ prot    : chr  "NO" "NO" "NO" "NO" ...
##  $ imm     : chr  "NO" "NO" "NO" "SI" ...
##  $ rpcrate : num  0.247 0.015 0.21 0.009 0.069 0.086 0.182 0.063 0.179 0.042 ..
.
##  $ prorate : num  0.263 0 0.336 0 0.289 0.434 0.583 0 0.172 0.167 ...

#Transformation of binary categorical variables
dati$prot_bin <- ifelse(dati$prot == "SI", 1, 0)
dati$imm_bin <- ifelse(dati$imm == "SI", 1, 0)

#Random splitting into training and validation sets
set.seed(123)
s <- sample.int(nrow(dati), 4000, replace = FALSE)

training <- dati[s, ]
validation <- dati[-s, ]
```

The command *set.seed(123)* sets a seed to ensure the reproducibility of the random sampling process. A random sample of 4,000 observations is selected for the training set, while the remaining 1,276 observations comprise the validation set.

The creation of univariate models (for each variable) is used as a preliminary step to assess each variable individually and determine whether it has a significant relationship with the target variable Flag.

```
#Univariate model for Age
mod_age <- lm(Flag ~ Age, data = training)
summary(mod_age)

##
## Call:
## lm(formula = Flag ~ Age, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5888 -0.5484  0.4308  0.4516  0.4749
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.4883627  0.0405817  12.034   <2e-16 ***
## Age         0.0012247  0.0007723   1.586    0.113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4973 on 3998 degrees of freedom
## Multiple R-squared:  0.0006287,  Adjusted R-squared:  0.0003787
## F-statistic: 2.515 on 1 and 3998 DF,  p-value: 0.1129
```

The variable Age is not statistically significant (p-value ≈ 0.113), indicating there is no strong evidence that Age alone has an impact on Flag.

```
#univariate model for zres
mod_zres <- lm(Flag ~ zres, data = training)
summary(mod_zres)

##
## Call:
## lm(formula = Flag ~ zres, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5831 -0.5606  0.4169  0.4394  0.4936
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.56063    0.01358  41.272  < 2e-16 ***
## zresNORD     0.02247    0.01896   1.185  0.23616
## zresSUD     -0.05426    0.01951  -2.781  0.00545 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.4965 on 3997 degrees of freedom
## Multiple R-squared:  0.004119,   Adjusted R-squared:  0.003621
## F-statistic: 8.266 on 2 and 3997 DF,  p-value: 0.0002616
```

The univariate model for zres shows a statistically significant effect for the category SUD, which is associated with a lower probability of Flag = 1 compared to the reference category (CENTRO). The NORD region, on the other hand, does not show a significant difference.

```
#univariate model for lastsal
mod_lastsal <- lm(Flag ~ lastsal, data = training)
summary(mod_lastsal)

##
## Call:
## lm(formula = Flag ~ lastsal, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5972 -0.5430  0.4088  0.4479  0.9723
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.997e-01  1.413e-02  42.435  < 2e-16 ***
## lastsal     -1.665e-05  4.057e-06  -4.103 4.16e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4964 on 3998 degrees of freedom
## Multiple R-squared:  0.004193,   Adjusted R-squared:  0.003944
## F-statistic: 16.84 on 1 and 3998 DF,  p-value: 4.158e-05
```

The univariate model for lastsal shows a statistically significant negative relationship with Flag (p-vlaue < 0.001).

```
#univariate model for lastrat
mod_lastrat <- lm(Flag ~ lastrat, data = training)
summary(mod_lastrat)

##
## Call:
## lm(formula = Flag ~ lastrat, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.6718 -0.5384  0.3392  0.4137  1.1354
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.781e-01  1.100e-02   61.63   <2e-16 ***
## lastrat     -6.921e-04  4.336e-05  -15.96   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4823 on 3998 degrees of freedom
## Multiple R-squared:  0.05991,    Adjusted R-squared:  0.05967
## F-statistic: 254.8 on 1 and 3998 DF,  p-value: < 2.2e-16
```

The univariate model for lastrat reveals a highly significant negative association with Flag. As lastrat increases, the probability of Flag = 1 decreases. The effect is more pronounced than in previous models.

```
#univariate model for avgsal
mod_avgsal <- lm(Flag ~ avgsal, data = training)
summary(mod_avgsal)

##
## Call:
## lm(formula = Flag ~ avgsal, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5732 -0.5507  0.4323  0.4473  0.7502
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.736e-01  1.146e-02  50.076  < 2e-16 ***
## avgsal      -6.771e-06  2.551e-06  -2.654  0.00798 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.497 on 3998 degrees of freedom
## Multiple R-squared:  0.001759,   Adjusted R-squared:  0.001509
## F-statistic: 7.045 on 1 and 3998 DF,  p-value: 0.007979
```

The univariate model for avgsal shows a statistically significant negative relationship with Flag ($p \approx 0.008$). This suggests that higher average salaries are modestly associated with a lower probability of Flag = 1.

```
#univariate model for recvsaff
mod_recvsaff <- lm(Flag ~ recvsaff, data = training)
summary(mod_recvsaff)

##
## Call:
## lm(formula = Flag ~ recvsaff, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1706 -0.5236  0.3886  0.4639  0.4868
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.513113   0.008382   61.22   <2e-16 ***
```

```
## recvsaff    0.026088   0.002201   11.85   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.489 on 3998 degrees of freedom
## Multiple R-squared:  0.03394,    Adjusted R-squared:  0.03369
## F-statistic: 140.4 on 1 and 3998 DF,  p-value: < 2.2e-16
```

The univariate model for recvsaff shows a highly significant positive association with Flag. The coefficient (≈ 0.026) indicates that when the ratio between repaid amount and total loan increases, the probability of Flag = 1 increases as well.

```
#univariate model for kept
mod_kept <- lm(Flag ~ kept, data = training)
summary(mod_kept)

##
## Call:
## lm(formula = Flag ~ kept, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.8247 -0.4578  0.1914  0.3594  0.8351
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.16485    0.01585   10.40   <2e-16 ***
## kept         0.65990    0.02409   27.39   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4565 on 3998 degrees of freedom
## Multiple R-squared:  0.158,  Adjusted R-squared:  0.1578
## F-statistic: 750.3 on 1 and 3998 DF,  p-value: < 2.2e-16
```

The univariate model for kept reveals a very strong and highly significant relationship with the target variable Flag. The coefficient of approximately 0.66 suggests that the ratio between the late repayments andf the total number of payments is associated with a substantially higher probability of Flag = 1.

```
#univariate model for prot
mod_prot <- lm(Flag ~ prot, data = training)
summary(mod_prot)

##
## Call:
## lm(formula = Flag ~ prot, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5518 -0.5518  0.4481  0.4481  0.4667
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.55185    0.00794  69.499   <2e-16 ***
## protSI      -0.01851    0.05799  -0.319     0.75
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4975 on 3998 degrees of freedom
## Multiple R-squared:  2.55e-05,   Adjusted R-squared:  -0.0002246
## F-statistic: 0.1019 on 1 and 3998 DF,  p-value: 0.7495
```

The univariate model for prot indicates no significant relationship with the outcome variable Flag. The p-value for protSI is 0.75, far above the standard significance thresholds, suggesting that the presence of "prot" does not have an impact on the Flag.

```
#univariate model for imm
mod_imm <- lm(Flag ~ imm, data = training)
summary(mod_imm)

##
## Call:
## lm(formula = Flag ~ imm, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5618 -0.5502  0.4382  0.4498  0.4498
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.550211   0.008343  65.947   <2e-16 ***
## immSI       0.011587   0.025014   0.463    0.643
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4975 on 3998 degrees of freedom
## Multiple R-squared:  5.367e-05,  Adjusted R-squared:  -0.0001964
## F-statistic: 0.2146 on 1 and 3998 DF,  p-value: 0.6432
```

The univariate model for imm shows no significant effect on the outcome Flag. The p-value for immSI is 0.643, indicating no statistically meaningful association.

```
#univariate model for rpcrate
mod_rpcrate <- lm(Flag ~ rpcrate, data = training)
summary(mod_rpcrate)

##
## Call:
## lm(formula = Flag ~ rpcrate, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.2474 -0.4161  0.1449  0.4283  0.6845
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.31545    0.01197   26.35   <2e-16 ***
## rpcrate      0.93198    0.03741   24.91   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4628 on 3998 degrees of freedom
## Multiple R-squared:  0.1344, Adjusted R-squared:  0.1342
## F-statistic: 620.6 on 1 and 3998 DF,  p-value: < 2.2e-16
```

The univariate model for rpcrate indicates a strong and highly significant positive association with the outcome Flag. The coefficient for rpcrate is 0.93198, meaning that as rpcrate increases, the likelihood of Flag being 1 increases substantially. The p-value is extremely low, confirming the statistical significance of this predictor.

```
#univariate model for prorate
mod_prorate <- lm(Flag ~ prorate, data = training)
summary(mod_prorate)

##
## Call:
## lm(formula = Flag ~ prorate, data = training)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.9574 -0.5224  0.3310  0.4352  0.6312
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.36880    0.01572   23.46   <2e-16 ***
## prorate      0.58858    0.04416   13.33   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4868 on 3998 degrees of freedom
## Multiple R-squared:  0.04254,    Adjusted R-squared:  0.04231
## F-statistic: 177.7 on 1 and 3998 DF,  p-value: < 2.2e-16
```

The model for prorate reveals a strong, statistically significant positive association with Flag. The very low p-value confirms this effect is highly significant. A coefficient of 0.589 indicates that higher prorate values increase the likelihood of Flag = 1.

In conclusion, these analyses indicate that variables related to financial status and repayment behavior, like: lastsal, lastrat, recvsaff, avgsal, rpcrate, prorate, and kept are significantly associated with the outcome variable Flag. These findings suggest that borrowers' financial obligations and their repayment discipline play a key role in explaining the variation in default risk. Moreover, the geographic variable zres shows a strong effect, particularly for individuals residing in the SUD area compared to those in CENTRO, highlighting potential regional disparities in default behavior. On the other hand: prot, imm, and Age do not exhibit a statistically significant effect when analyzed independently.

c) From the covariate set identified at point b) fit a large model M1 as a hierarchical model that includes all possible second order interaction terms that involves at least one categorical/binary covariate. Starting from this M1 model, develop a backward selection procedure to arrive at the final model.

```
M1 <- lm(Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
           prot_bin + imm_bin + rpcrate + prorate +
           Age * zres + Age * prot_bin + Age * imm_bin +
           zres * prot_bin + zres * imm_bin +
           lastsal * zres + lastsal * prot_bin + lastsal * imm_bin +
           lastrat * zres + lastrat * prot_bin + lastrat * imm_bin +
           avgsal * zres + avgsal * prot_bin + avgsal * imm_bin +
           recvsaff * zres + recvsaff * prot_bin + recvsaff * imm_bin +
           kept * zres + kept * prot_bin + kept * imm_bin +
           rpcrate * zres + rpcrate * prot_bin + rpcrate * imm_bin +
           prorate * zres + prorate * prot_bin + prorate * imm_bin +
           prot_bin * imm_bin,
         data = dati)

#backward elimination
mod.fin<-stepAIC(M1,direction="backward", test="Chisq")

## Start:  AIC=-8892.32
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age * zres + Age *
##     prot_bin + Age * imm_bin + zres * prot_bin + zres * imm_bin +
##     lastsal * zres + lastsal * prot_bin + lastsal * imm_bin +
##     lastrat * zres + lastrat * prot_bin + lastrat * imm_bin +
##     avgsal * zres + avgsal * prot_bin + avgsal * imm_bin + recvsaff *
##     zres + recvsaff * prot_bin + recvsaff * imm_bin + kept *
##     zres + kept * prot_bin + kept * imm_bin + rpcrate * zres +
##     rpcrate * prot_bin + rpcrate * imm_bin + prorate * zres +
##     prorate * prot_bin + prorate * imm_bin + prot_bin * imm_bin
##
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - zres:prorate        2   0.02456 959.65 -8896.2 < 2.2e-16 ***
## - zres:imm_bin        2   0.06119 959.68 -8896.0 < 2.2e-16 ***
## - zres:lastrat        2   0.08814 959.71 -8895.8 < 2.2e-16 ***
## - zres:lastsal        2   0.29220 959.92 -8894.7 < 2.2e-16 ***
## - prot_bin:imm_bin    1   0.00115 959.62 -8894.3 < 2.2e-16 ***
## - avgsal:prot_bin     1   0.00152 959.62 -8894.3 < 2.2e-16 ***
## - avgsal:imm_bin      1   0.00207 959.63 -8894.3 < 2.2e-16 ***
## - lastrat:imm_bin     1   0.00469 959.63 -8894.3 < 2.2e-16 ***
## - imm_bin:prorate     1   0.03689 959.66 -8894.1 < 2.2e-16 ***
## - Age:prot_bin        1   0.05226 959.68 -8894.0 < 2.2e-16 ***
## - zres:prot_bin       2   0.43245 960.06 -8893.9 < 2.2e-16 ***
## - kept:prot_bin       1   0.07498 959.70 -8893.9 < 2.2e-16 ***
## - Age:zres            2   0.44165 960.06 -8893.9 < 2.2e-16 ***
## - zres:rpcrate        2   0.44443 960.07 -8893.9 < 2.2e-16 ***
## - lastsal:prot_bin    1   0.09538 959.72 -8893.8 < 2.2e-16 ***
## - prot_bin:rpcrate    1   0.10956 959.73 -8893.7 < 2.2e-16 ***
```

```
## - prot_bin:prorate   1   0.16435 959.79 -8893.4 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.28889 959.91 -8892.7 < 2.2e-16 ***
## - kept:imm_bin       1   0.33950 959.96 -8892.5 < 2.2e-16 ***
## <none>                           959.62 -8892.3
## - zres:avgsal        2   0.74182 960.36 -8892.2 < 2.2e-16 ***
## - zres:kept          2   0.82254 960.45 -8891.8 < 2.2e-16 ***
## - zres:recvsaff      2   0.84531 960.47 -8891.7 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.64887 960.27 -8890.8 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.67072 960.29 -8890.6 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.67267 960.30 -8890.6 < 2.2e-16 ***
## - Age:imm_bin        1   0.83653 960.46 -8889.7 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.48123 961.10 -8886.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8896.19
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + zres:imm_bin + zres:lastsal +
##     lastsal:prot_bin + lastsal:imm_bin + zres:lastrat + lastrat:prot_bin +
##     lastrat:imm_bin + zres:avgsal + avgsal:prot_bin + avgsal:imm_bin +
##     zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin + zres:kept +
##     kept:prot_bin + kept:imm_bin + zres:rpcrate + prot_bin:rpcrate +
##     imm_bin:rpcrate + prot_bin:prorate + imm_bin:prorate + prot_bin:imm_bin
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - zres:imm_bin       2   0.05774 959.71 -8899.9 < 2.2e-16 ***
## - zres:lastrat       2   0.09065 959.74 -8899.7 < 2.2e-16 ***
## - zres:lastsal       2   0.29134 959.94 -8898.6 < 2.2e-16 ***
## - prot_bin:imm_bin   1   0.00130 959.65 -8898.2 < 2.2e-16 ***
## - avgsal:prot_bin    1   0.00142 959.65 -8898.2 < 2.2e-16 ***
## - avgsal:imm_bin     1   0.00219 959.65 -8898.2 < 2.2e-16 ***
## - lastrat:imm_bin    1   0.00474 959.65 -8898.2 < 2.2e-16 ***
## - imm_bin:prorate    1   0.03654 959.68 -8898.0 < 2.2e-16 ***
## - Age:prot_bin       1   0.05225 959.70 -8897.9 < 2.2e-16 ***
## - zres:prot_bin      2   0.43173 960.08 -8897.8 < 2.2e-16 ***
## - kept:prot_bin      1   0.07420 959.72 -8897.8 < 2.2e-16 ***
## - zres:rpcrate       2   0.44664 960.09 -8897.7 < 2.2e-16 ***
## - Age:zres           2   0.45733 960.11 -8897.7 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.09494 959.74 -8897.7 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.10878 959.76 -8897.6 < 2.2e-16 ***
## - prot_bin:prorate   1   0.16474 959.81 -8897.3 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.28912 959.94 -8896.6 < 2.2e-16 ***
## - kept:imm_bin       1   0.33937 959.99 -8896.3 < 2.2e-16 ***
## <none>                           959.65 -8896.2
## - zres:avgsal        2   0.74349 960.39 -8896.1 < 2.2e-16 ***
## - zres:recvsaff      2   0.86927 960.52 -8895.4 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.64688 960.29 -8894.6 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.66958 960.32 -8894.5 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.67226 960.32 -8894.5 < 2.2e-16 ***
```

```
## - zres:kept          2   1.14820 960.80 -8893.9 < 2.2e-16 ***
## - Age:imm_bin        1   0.84013 960.49 -8893.6 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.48905 961.14 -8890.0 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8899.87
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + zres:lastsal + lastsal:prot_bin +
##     lastsal:imm_bin + zres:lastrat + lastrat:prot_bin + lastrat:imm_bin +
##     zres:avgsal + avgsal:prot_bin + avgsal:imm_bin + zres:recvsaff +
##     recvsaff:prot_bin + recvsaff:imm_bin + zres:kept + kept:prot_bin +
##     kept:imm_bin + zres:rpcrate + prot_bin:rpcrate + imm_bin:rpcrate +
##     prot_bin:prorate + imm_bin:prorate + prot_bin:imm_bin
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - zres:lastrat       2   0.09256 959.80 -8903.4 < 2.2e-16 ***
## - zres:lastsal       2   0.29076 960.00 -8902.3 < 2.2e-16 ***
## - prot_bin:imm_bin   1   0.00137 959.71 -8901.9 < 2.2e-16 ***
## - avgsal:imm_bin     1   0.00246 959.71 -8901.9 < 2.2e-16 ***
## - avgsal:prot_bin    1   0.00257 959.71 -8901.9 < 2.2e-16 ***
## - lastrat:imm_bin    1   0.00556 959.71 -8901.8 < 2.2e-16 ***
## - imm_bin:prorate    1   0.04014 959.75 -8901.7 < 2.2e-16 ***
## - Age:prot_bin       1   0.05484 959.76 -8901.6 < 2.2e-16 ***
## - kept:prot_bin      1   0.07304 959.78 -8901.5 < 2.2e-16 ***
## - zres:rpcrate       2   0.44253 960.15 -8901.4 < 2.2e-16 ***
## - zres:prot_bin      2   0.44861 960.15 -8901.4 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.09030 959.80 -8901.4 < 2.2e-16 ***
## - Age:zres           2   0.46354 960.17 -8901.3 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.10513 959.81 -8901.3 < 2.2e-16 ***
## - prot_bin:prorate   1   0.16304 959.87 -8901.0 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29253 960.00 -8900.3 < 2.2e-16 ***
## - kept:imm_bin       1   0.34103 960.05 -8900.0 < 2.2e-16 ***
## <none>                           959.71 -8899.9
## - zres:avgsal        2   0.77896 960.48 -8899.6 < 2.2e-16 ***
## - zres:recvsaff      2   0.84848 960.55 -8899.2 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.65098 960.36 -8898.3 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.69310 960.40 -8898.1 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.69792 960.40 -8898.0 < 2.2e-16 ***
## - zres:kept          2   1.13656 960.84 -8897.6 < 2.2e-16 ***
## - Age:imm_bin        1   0.83940 960.54 -8897.3 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.47464 961.18 -8893.8 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8903.36
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + zres:lastsal + lastsal:prot_bin +
```

```
##     lastsal:imm_bin + lastrat:prot_bin + lastrat:imm_bin + zres:avgsal +
##     avgsal:prot_bin + avgsal:imm_bin + zres:recvsaff + recvsaff:prot_bin +
##     recvsaff:imm_bin + zres:kept + kept:prot_bin + kept:imm_bin +
##     zres:rpcrate + prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate +
##     imm_bin:prorate + prot_bin:imm_bin
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - zres:lastsal       2   0.27183 960.07 -8905.9 < 2.2e-16 ***
## - prot_bin:imm_bin   1   0.00124 959.80 -8905.4 < 2.2e-16 ***
## - avgsal:imm_bin     1   0.00175 959.80 -8905.4 < 2.2e-16 ***
## - avgsal:prot_bin    1   0.00334 959.80 -8905.3 < 2.2e-16 ***
## - lastrat:imm_bin    1   0.00659 959.80 -8905.3 < 2.2e-16 ***
## - zres:rpcrate       2   0.39591 960.19 -8905.2 < 2.2e-16 ***
## - imm_bin:prorate    1   0.04154 959.84 -8905.1 < 2.2e-16 ***
## - Age:prot_bin       1   0.05742 959.86 -8905.0 < 2.2e-16 ***
## - kept:prot_bin      1   0.07353 959.87 -8905.0 < 2.2e-16 ***
## - zres:prot_bin      2   0.44606 960.24 -8904.9 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.09385 959.89 -8904.8 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.10452 959.90 -8904.8 < 2.2e-16 ***
## - Age:zres           2   0.47539 960.27 -8904.8 < 2.2e-16 ***
## - prot_bin:prorate   1   0.16134 959.96 -8904.5 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29232 960.09 -8903.8 < 2.2e-16 ***
## - kept:imm_bin       1   0.34410 960.14 -8903.5 < 2.2e-16 ***
## <none>                           959.80 -8903.4
## - zres:avgsal        2   0.78123 960.58 -8903.1 < 2.2e-16 ***
## - zres:recvsaff      2   0.80432 960.60 -8902.9 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.65952 960.46 -8901.7 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.69129 960.49 -8901.6 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.69142 960.49 -8901.6 < 2.2e-16 ***
## - zres:kept          2   1.19087 960.99 -8900.8 < 2.2e-16 ***
## - Age:imm_bin        1   0.85443 960.65 -8900.7 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.48017 961.28 -8897.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8905.87
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + lastrat:imm_bin + zres:avgsal + avgsal:prot_bin +
##     avgsal:imm_bin + zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin +
##     zres:kept + kept:prot_bin + kept:imm_bin + zres:rpcrate +
##     prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate + imm_bin:prorate +
##     prot_bin:imm_bin
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - prot_bin:imm_bin   1   0.00102 960.07 -8907.9 < 2.2e-16 ***
## - avgsal:prot_bin    1   0.00110 960.07 -8907.9 < 2.2e-16 ***
## - lastrat:imm_bin    1   0.00537 960.08 -8907.8 < 2.2e-16 ***
## - avgsal:imm_bin     1   0.00615 960.08 -8907.8 < 2.2e-16 ***
```

```
## - imm_bin:prorate    1   0.03240 960.10 -8907.7 < 2.2e-16 ***
## - zres:rpcrate       2   0.40095 960.47 -8907.7 < 2.2e-16 ***
## - Age:prot_bin       1   0.05852 960.13 -8907.5 < 2.2e-16 ***
## - kept:prot_bin      1   0.08285 960.15 -8907.4 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.08875 960.16 -8907.4 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.10878 960.18 -8907.3 < 2.2e-16 ***
## - Age:zres           2   0.47444 960.54 -8907.3 < 2.2e-16 ***
## - zres:prot_bin      2   0.50648 960.58 -8907.1 < 2.2e-16 ***
## - prot_bin:prorate   1   0.17353 960.24 -8906.9 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29443 960.36 -8906.3 < 2.2e-16 ***
## - zres:avgsal        2   0.67272 960.74 -8906.2 < 2.2e-16 ***
## - kept:imm_bin       1   0.35165 960.42 -8905.9 < 2.2e-16 ***
## <none>                           960.07 -8905.9
## - zres:recvsaff      2   0.83736 960.91 -8905.3 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.66640 960.74 -8904.2 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.70451 960.77 -8904.0 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.71943 960.79 -8903.9 < 2.2e-16 ***
## - Age:imm_bin        1   0.85057 960.92 -8903.2 < 2.2e-16 ***
## - zres:kept          2   1.22893 961.30 -8903.1 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.47179 961.54 -8899.8 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8907.86
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + lastrat:imm_bin + zres:avgsal + avgsal:prot_bin +
##     avgsal:imm_bin + zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin +
##     zres:kept + kept:prot_bin + kept:imm_bin + zres:rpcrate +
##     prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate + imm_bin:prorate
##
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - avgsal:prot_bin    1   0.00144 960.07 -8909.9 < 2.2e-16 ***
## - lastrat:imm_bin    1   0.00532 960.08 -8909.8 < 2.2e-16 ***
## - avgsal:imm_bin     1   0.00621 960.08 -8909.8 < 2.2e-16 ***
## - imm_bin:prorate    1   0.03315 960.10 -8909.7 < 2.2e-16 ***
## - zres:rpcrate       2   0.40085 960.47 -8909.7 < 2.2e-16 ***
## - Age:prot_bin       1   0.06312 960.13 -8909.5 < 2.2e-16 ***
## - kept:prot_bin      1   0.08301 960.15 -8909.4 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.08841 960.16 -8909.4 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.10846 960.18 -8909.3 < 2.2e-16 ***
## - Age:zres           2   0.47435 960.55 -8909.3 < 2.2e-16 ***
## - zres:prot_bin      2   0.50665 960.58 -8909.1 < 2.2e-16 ***
## - prot_bin:prorate   1   0.17923 960.25 -8908.9 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29341 960.36 -8908.3 < 2.2e-16 ***
## - zres:avgsal        2   0.67346 960.74 -8908.2 < 2.2e-16 ***
## - kept:imm_bin       1   0.35070 960.42 -8907.9 < 2.2e-16 ***
## <none>                           960.07 -8907.9
## - zres:recvsaff      2   0.83813 960.91 -8907.3 < 2.2e-16 ***
```

```
## - lastrat:prot_bin   1   0.67549 960.75 -8906.2 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.70642 960.78 -8906.0 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.72004 960.79 -8905.9 < 2.2e-16 ***
## - Age:imm_bin        1   0.84992 960.92 -8905.2 < 2.2e-16 ***
## - zres:kept          2   1.22870 961.30 -8905.1 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.47231 961.54 -8901.8 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8909.86
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + lastrat:imm_bin + zres:avgsal + avgsal:imm_bin +
##     zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin + zres:kept +
##     kept:prot_bin + kept:imm_bin + zres:rpcrate + prot_bin:rpcrate +
##     imm_bin:rpcrate + prot_bin:prorate + imm_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - lastrat:imm_bin    1   0.00521 960.08 -8911.8 < 2.2e-16 ***
## - avgsal:imm_bin     1   0.01223 960.08 -8911.8 < 2.2e-16 ***
## - imm_bin:prorate    1   0.03307 960.11 -8911.7 < 2.2e-16 ***
## - zres:rpcrate       2   0.40103 960.47 -8911.7 < 2.2e-16 ***
## - Age:prot_bin       1   0.06286 960.14 -8911.5 < 2.2e-16 ***
## - kept:prot_bin      1   0.08360 960.16 -8911.4 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.09280 960.17 -8911.3 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.10979 960.18 -8911.3 < 2.2e-16 ***
## - Age:zres           2   0.47496 960.55 -8911.2 < 2.2e-16 ***
## - zres:prot_bin      2   0.50622 960.58 -8911.1 < 2.2e-16 ***
## - prot_bin:prorate   1   0.17846 960.25 -8910.9 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29210 960.36 -8910.3 < 2.2e-16 ***
## - kept:imm_bin       1   0.34972 960.42 -8909.9 < 2.2e-16 ***
## - zres:avgsal        2   0.72742 960.80 -8909.9 < 2.2e-16 ***
## <none>                           960.07 -8909.9
## - zres:recvsaff      2   0.83670 960.91 -8909.3 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.67407 960.75 -8908.2 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.70577 960.78 -8908.0 < 2.2e-16 ***
## - lastsal:imm_bin    1   0.73105 960.80 -8907.8 < 2.2e-16 ***
## - Age:imm_bin        1   0.84892 960.92 -8907.2 < 2.2e-16 ***
## - zres:kept          2   1.22762 961.30 -8907.1 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.47234 961.54 -8903.8 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8911.83
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + zres:avgsal + avgsal:imm_bin + zres:recvsaff +
##     recvsaff:prot_bin + recvsaff:imm_bin + zres:kept + kept:prot_bin +
```

```
##     kept:imm_bin + zres:rpcrate + prot_bin:rpcrate + imm_bin:rpcrate +
##     prot_bin:prorate + imm_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - avgsal:imm_bin     1   0.01244 960.09 -8913.8 < 2.2e-16 ***
## - imm_bin:prorate    1   0.03120 960.11 -8913.7 < 2.2e-16 ***
## - zres:rpcrate       2   0.40145 960.48 -8913.6 < 2.2e-16 ***
## - Age:prot_bin       1   0.06207 960.14 -8913.5 < 2.2e-16 ***
## - kept:prot_bin      1   0.08315 960.16 -8913.4 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.08817 960.17 -8913.3 < 2.2e-16 ***
## - Age:zres           2   0.47310 960.55 -8913.2 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.11204 960.19 -8913.2 < 2.2e-16 ***
## - zres:prot_bin      2   0.50488 960.58 -8913.1 < 2.2e-16 ***
## - prot_bin:prorate   1   0.18002 960.26 -8912.8 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29405 960.37 -8912.2 < 2.2e-16 ***
## - kept:imm_bin       1   0.34517 960.42 -8911.9 < 2.2e-16 ***
## - zres:avgsal        2   0.72632 960.80 -8911.8 < 2.2e-16 ***
## <none>                           960.08 -8911.8
## - zres:recvsaff      2   0.83678 960.91 -8911.2 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.67661 960.75 -8910.1 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.71906 960.80 -8909.9 < 2.2e-16 ***
## - Age:imm_bin        1   0.84625 960.92 -8909.2 < 2.2e-16 ***
## - zres:kept          2   1.22524 961.30 -8909.1 < 2.2e-16 ***
## - lastsal:imm_bin    1   1.22010 961.30 -8907.1 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.47168 961.55 -8905.7 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8913.76
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + zres:avgsal + zres:recvsaff + recvsaff:prot_bin +
##     recvsaff:imm_bin + zres:kept + kept:prot_bin + kept:imm_bin +
##     zres:rpcrate + prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate +
##     imm_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - imm_bin:prorate    1   0.03067 960.12 -8915.6 < 2.2e-16 ***
## - zres:rpcrate       2   0.39981 960.49 -8915.6 < 2.2e-16 ***
## - Age:prot_bin       1   0.06018 960.15 -8915.4 < 2.2e-16 ***
## - kept:prot_bin      1   0.08354 960.17 -8915.3 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.08966 960.18 -8915.3 < 2.2e-16 ***
## - Age:zres           2   0.47501 960.56 -8915.1 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.11383 960.20 -8915.1 < 2.2e-16 ***
## - zres:prot_bin      2   0.49826 960.59 -8915.0 < 2.2e-16 ***
## - prot_bin:prorate   1   0.17631 960.27 -8914.8 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29299 960.38 -8914.1 < 2.2e-16 ***
## - kept:imm_bin       1   0.34731 960.44 -8913.8 < 2.2e-16 ***
## - zres:avgsal        2   0.71769 960.81 -8913.8 < 2.2e-16 ***
```

```
## <none>                             960.09 -8913.8
## - zres:recvsaff      2   0.83187 960.92 -8913.2 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.67501 960.76 -8912.1 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.72803 960.82 -8911.8 < 2.2e-16 ***
## - Age:imm_bin        1   0.85034 960.94 -8911.1 < 2.2e-16 ***
## - zres:kept          2   1.22282 961.31 -8911.0 < 2.2e-16 ***
## - lastsal:imm_bin    1   1.26651 961.36 -8908.8 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.47261 961.56 -8907.7 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8915.59
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + zres:avgsal + zres:recvsaff + recvsaff:prot_bin +
##     recvsaff:imm_bin + zres:kept + kept:prot_bin + kept:imm_bin +
##     zres:rpcrate + prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - zres:rpcrate       2   0.39503 960.52 -8917.4 < 2.2e-16 ***
## - Age:prot_bin       1   0.05984 960.18 -8917.3 < 2.2e-16 ***
## - lastsal:prot_bin   1   0.09071 960.21 -8917.1 < 2.2e-16 ***
## - kept:prot_bin      1   0.09544 960.22 -8917.1 < 2.2e-16 ***
## - Age:zres           2   0.48003 960.60 -8917.0 < 2.2e-16 ***
## - prot_bin:rpcrate   1   0.11697 960.24 -8916.9 < 2.2e-16 ***
## - zres:prot_bin      2   0.50249 960.62 -8916.8 < 2.2e-16 ***
## - prot_bin:prorate   1   0.23349 960.35 -8916.3 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.29747 960.42 -8916.0 < 2.2e-16 ***
## - zres:avgsal        2   0.71054 960.83 -8915.7 < 2.2e-16 ***
## <none>                           960.12 -8915.6
## - zres:recvsaff      2   0.82990 960.95 -8915.0 < 2.2e-16 ***
## - kept:imm_bin       1   0.49992 960.62 -8914.8 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.67361 960.79 -8913.9 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.72941 960.85 -8913.6 < 2.2e-16 ***
## - zres:kept          2   1.21385 961.33 -8912.9 < 2.2e-16 ***
## - Age:imm_bin        1   0.89000 961.01 -8912.7 < 2.2e-16 ***
## - lastsal:imm_bin    1   1.33439 961.45 -8910.3 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.46891 961.59 -8909.5 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8917.42
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:prot_bin +
##     Age:imm_bin + zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin +
##     lastrat:prot_bin + zres:avgsal + zres:recvsaff + recvsaff:prot_bin +
##     recvsaff:imm_bin + zres:kept + kept:prot_bin + kept:imm_bin +
##     prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate
##
```

```
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - Age:prot_bin        1   0.05889 960.57 -8919.1 < 2.2e-16 ***
## - lastsal:prot_bin    1   0.09023 960.61 -8918.9 < 2.2e-16 ***
## - kept:prot_bin       1   0.10397 960.62 -8918.8 < 2.2e-16 ***
## - prot_bin:rpcrate    1   0.12706 960.64 -8918.7 < 2.2e-16 ***
## - zres:prot_bin       2   0.49776 961.01 -8918.7 < 2.2e-16 ***
## - Age:zres            2   0.55295 961.07 -8918.4 < 2.2e-16 ***
## - prot_bin:prorate    1   0.23046 960.75 -8918.2 < 2.2e-16 ***
## - recvsaff:prot_bin   1   0.29699 960.81 -8917.8 < 2.2e-16 ***
## - zres:avgsal         2   0.71090 961.23 -8917.5 < 2.2e-16 ***
## <none>                            960.52 -8917.4
## - zres:kept           2   0.84466 961.36 -8916.8 < 2.2e-16 ***
## - kept:imm_bin        1   0.49302 961.01 -8916.7 < 2.2e-16 ***
## - zres:recvsaff       2   0.91983 961.44 -8916.4 < 2.2e-16 ***
## - lastrat:prot_bin    1   0.67119 961.19 -8915.7 < 2.2e-16 ***
## - recvsaff:imm_bin    1   0.74117 961.26 -8915.3 < 2.2e-16 ***
## - Age:imm_bin         1   0.86743 961.38 -8914.7 < 2.2e-16 ***
## - lastsal:imm_bin     1   1.34208 961.86 -8912.1 < 2.2e-16 ***
## - imm_bin:rpcrate     1   1.50295 962.02 -8911.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8919.1
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:imm_bin +
##     zres:prot_bin + lastsal:prot_bin + lastsal:imm_bin + lastrat:prot_bin +
##     zres:avgsal + zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin +
##     zres:kept + kept:prot_bin + kept:imm_bin + prot_bin:rpcrate +
##     imm_bin:rpcrate + prot_bin:prorate
##
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - lastsal:prot_bin    1   0.08537 960.66 -8920.6 < 2.2e-16 ***
## - kept:prot_bin       1   0.09901 960.67 -8920.6 < 2.2e-16 ***
## - zres:prot_bin       2   0.51372 961.09 -8920.3 < 2.2e-16 ***
## - prot_bin:rpcrate    1   0.17176 960.75 -8920.2 < 2.2e-16 ***
## - prot_bin:prorate    1   0.19012 960.76 -8920.1 < 2.2e-16 ***
## - Age:zres            2   0.56887 961.14 -8920.0 < 2.2e-16 ***
## - recvsaff:prot_bin   1   0.28130 960.86 -8919.6 < 2.2e-16 ***
## - zres:avgsal         2   0.72255 961.30 -8919.1 < 2.2e-16 ***
## <none>                            960.57 -8919.1
## - zres:kept           2   0.83709 961.41 -8918.5 < 2.2e-16 ***
## - kept:imm_bin        1   0.50080 961.08 -8918.3 < 2.2e-16 ***
## - zres:recvsaff       2   0.92176 961.50 -8918.0 < 2.2e-16 ***
## - lastrat:prot_bin    1   0.67011 961.24 -8917.4 < 2.2e-16 ***
## - recvsaff:imm_bin    1   0.74539 961.32 -8917.0 < 2.2e-16 ***
## - Age:imm_bin         1   0.81922 961.39 -8916.6 < 2.2e-16 ***
## - lastsal:imm_bin     1   1.35533 961.93 -8913.7 < 2.2e-16 ***
## - imm_bin:rpcrate     1   1.49414 962.07 -8912.9 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Step:  AIC=-8920.63
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:imm_bin +
##     zres:prot_bin + lastsal:imm_bin + lastrat:prot_bin + zres:avgsal +
##     zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin + zres:kept +
##     kept:prot_bin + kept:imm_bin + prot_bin:rpcrate + imm_bin:rpcrate +
##     prot_bin:prorate
##
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - kept:prot_bin       1   0.12467 960.78 -8921.9 < 2.2e-16 ***
## - zres:prot_bin       2   0.52314 961.18 -8921.8 < 2.2e-16 ***
## - prot_bin:rpcrate    1   0.18351 960.84 -8921.6 < 2.2e-16 ***
## - Age:zres            2   0.56953 961.23 -8921.5 < 2.2e-16 ***
## - prot_bin:prorate    1   0.21657 960.88 -8921.4 < 2.2e-16 ***
## - zres:avgsal         2   0.69669 961.36 -8920.8 < 2.2e-16 ***
## <none>                            960.66 -8920.6
## - recvsaff:prot_bin   1   0.38669 961.05 -8920.5 < 2.2e-16 ***
## - zres:kept           2   0.83200 961.49 -8920.1 < 2.2e-16 ***
## - kept:imm_bin        1   0.49566 961.16 -8919.9 < 2.2e-16 ***
## - zres:recvsaff       2   0.91632 961.58 -8919.6 < 2.2e-16 ***
## - lastrat:prot_bin    1   0.73226 961.39 -8918.6 < 2.2e-16 ***
## - recvsaff:imm_bin    1   0.73871 961.40 -8918.6 < 2.2e-16 ***
## - Age:imm_bin         1   0.80781 961.47 -8918.2 < 2.2e-16 ***
## - lastsal:imm_bin     1   1.28118 961.94 -8915.6 < 2.2e-16 ***
## - imm_bin:rpcrate     1   1.47921 962.14 -8914.5 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8921.94
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:imm_bin +
##     zres:prot_bin + lastsal:imm_bin + lastrat:prot_bin + zres:avgsal +
##     zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin + zres:kept +
##     kept:imm_bin + prot_bin:rpcrate + imm_bin:rpcrate + prot_bin:prorate
##
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - prot_bin:rpcrate    1   0.07613 960.86 -8923.5 < 2.2e-16 ***
## - prot_bin:prorate    1   0.15017 960.93 -8923.1 < 2.2e-16 ***
## - zres:prot_bin       2   0.52679 961.31 -8923.1 < 2.2e-16 ***
## - Age:zres            2   0.56263 961.35 -8922.9 < 2.2e-16 ***
## - recvsaff:prot_bin   1   0.31125 961.10 -8922.2 < 2.2e-16 ***
## - zres:avgsal         2   0.69702 961.48 -8922.1 < 2.2e-16 ***
## <none>                            960.78 -8921.9
## - kept:imm_bin        1   0.43014 961.21 -8921.6 < 2.2e-16 ***
## - zres:kept           2   0.84479 961.63 -8921.3 < 2.2e-16 ***
## - zres:recvsaff       2   0.91870 961.70 -8920.9 < 2.2e-16 ***
## - recvsaff:imm_bin    1   0.71853 961.50 -8920.0 < 2.2e-16 ***
## - lastrat:prot_bin    1   0.74495 961.53 -8919.9 < 2.2e-16 ***
## - Age:imm_bin         1   0.83845 961.62 -8919.3 < 2.2e-16 ***
```

```
## - lastsal:imm_bin    1   1.26732 962.05 -8917.0 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.54886 962.33 -8915.4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8923.52
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:imm_bin +
##     zres:prot_bin + lastsal:imm_bin + lastrat:prot_bin + zres:avgsal +
##     zres:recvsaff + recvsaff:prot_bin + recvsaff:imm_bin + zres:kept +
##     kept:imm_bin + imm_bin:rpcrate + prot_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - zres:prot_bin      2   0.55186 961.41 -8924.5 < 2.2e-16 ***
## - Age:zres           2   0.56631 961.43 -8924.4 < 2.2e-16 ***
## - prot_bin:prorate   1   0.20341 961.06 -8924.4 < 2.2e-16 ***
## - zres:avgsal        2   0.71655 961.58 -8923.6 < 2.2e-16 ***
## <none>                           960.86 -8923.5
## - recvsaff:prot_bin  1   0.41757 961.28 -8923.2 < 2.2e-16 ***
## - kept:imm_bin       1   0.44488 961.31 -8923.1 < 2.2e-16 ***
## - zres:kept          2   0.84661 961.71 -8922.9 < 2.2e-16 ***
## - zres:recvsaff      2   0.91830 961.78 -8922.5 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.73826 961.60 -8921.5 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.79929 961.66 -8921.1 < 2.2e-16 ***
## - Age:imm_bin        1   0.81370 961.67 -8921.1 < 2.2e-16 ***
## - lastsal:imm_bin    1   1.27615 962.14 -8918.5 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.66783 962.53 -8916.4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8924.49
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:zres + Age:imm_bin +
##     lastsal:imm_bin + lastrat:prot_bin + zres:avgsal + zres:recvsaff +
##     recvsaff:prot_bin + recvsaff:imm_bin + zres:kept + kept:imm_bin +
##     imm_bin:rpcrate + prot_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## - Age:zres           2   0.59387 962.01 -8925.2 < 2.2e-16 ***
## - recvsaff:prot_bin  1   0.31452 961.73 -8924.8 < 2.2e-16 ***
## - prot_bin:prorate   1   0.32564 961.74 -8924.7 < 2.2e-16 ***
## <none>                           961.41 -8924.5
## - zres:avgsal        2   0.76512 962.18 -8924.3 < 2.2e-16 ***
## - kept:imm_bin       1   0.44742 961.86 -8924.0 < 2.2e-16 ***
## - zres:kept          2   0.84120 962.25 -8923.9 < 2.2e-16 ***
## - zres:recvsaff      2   0.90674 962.32 -8923.5 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.71672 962.13 -8922.6 < 2.2e-16 ***
## - Age:imm_bin        1   0.82069 962.23 -8922.0 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.84889 962.26 -8921.8 < 2.2e-16 ***
## - lastsal:imm_bin    1   1.32957 962.74 -8919.2 < 2.2e-16 ***
```

```
## - imm_bin:rpcrate    1   1.68843 963.10 -8917.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8925.24
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:imm_bin + lastsal:imm_bin +
##     lastrat:prot_bin + zres:avgsal + zres:recvsaff + recvsaff:prot_bin +
##     recvsaff:imm_bin + zres:kept + kept:imm_bin + imm_bin:rpcrate +
##     prot_bin:prorate
##
##                      Df Sum of Sq    RSS     AIC   Pr(Chi)
## - recvsaff:prot_bin  1   0.31029 962.32 -8925.5 < 2.2e-16 ***
## - prot_bin:prorate   1   0.33309 962.34 -8925.4 < 2.2e-16 ***
## <none>                           962.01 -8925.2
## - zres:avgsal        2   0.79429 962.80 -8924.9 < 2.2e-16 ***
## - kept:imm_bin       1   0.44820 962.45 -8924.8 < 2.2e-16 ***
## - zres:recvsaff      2   0.86273 962.87 -8924.5 < 2.2e-16 ***
## - zres:kept          2   0.88053 962.89 -8924.4 < 2.2e-16 ***
## - recvsaff:imm_bin   1   0.73321 962.74 -8923.2 < 2.2e-16 ***
## - Age:imm_bin        1   0.78195 962.79 -8923.0 < 2.2e-16 ***
## - lastrat:prot_bin   1   0.84320 962.85 -8922.6 < 2.2e-16 ***
## - lastsal:imm_bin    1   1.32030 963.33 -8920.0 < 2.2e-16 ***
## - imm_bin:rpcrate    1   1.70095 963.71 -8917.9 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Step:  AIC=-8925.54
## Flag ~ Age + zres + lastsal + lastrat + avgsal + recvsaff + kept +
##     prot_bin + imm_bin + rpcrate + prorate + Age:imm_bin + lastsal:imm_bin +
##     lastrat:prot_bin + zres:avgsal + zres:recvsaff + recvsaff:imm_bin +
##     zres:kept + kept:imm_bin + imm_bin:rpcrate + prot_bin:prorate
##
##                     Df Sum of Sq    RSS     AIC   Pr(Chi)
## <none>                           962.32 -8925.5
## - zres:avgsal       2   0.76388 963.08 -8925.3 < 2.2e-16 ***
## - prot_bin:prorate  1   0.41834 962.74 -8925.2 < 2.2e-16 ***
## - kept:imm_bin      1   0.45994 962.78 -8925.0 < 2.2e-16 ***
## - zres:recvsaff     2   0.87712 963.19 -8924.7 < 2.2e-16 ***
## - zres:kept         2   0.88305 963.20 -8924.7 < 2.2e-16 ***
## - recvsaff:imm_bin  1   0.65365 962.97 -8924.0 < 2.2e-16 ***
## - Age:imm_bin       1   0.78924 963.11 -8923.2 < 2.2e-16 ***
## - lastrat:prot_bin  1   0.88665 963.20 -8922.7 < 2.2e-16 ***
## - lastsal:imm_bin   1   1.29678 963.61 -8920.4 < 2.2e-16 ***
## - imm_bin:rpcrate   1   1.70776 964.02 -8918.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

mod.fin
```

```
##
## Call:
## lm(formula = Flag ~ Age + zres + lastsal + lastrat + avgsal +
##     recvsaff + kept + prot_bin + imm_bin + rpcrate + prorate +
##     Age:imm_bin + lastsal:imm_bin + lastrat:prot_bin + zres:avgsal +
##     zres:recvsaff + recvsaff:imm_bin + zres:kept + kept:imm_bin +
##     imm_bin:rpcrate + prot_bin:prorate, data = dati)
##
## Coefficients:
##       (Intercept)                Age           zresNORD             zresSUD
##         1.471e-01          8.096e-04          1.436e-02         -1.265e-02
##           lastsal            lastrat             avgsal            recvsaff
##         2.481e-05         -6.475e-04          1.706e-06          2.283e-02
##               kept           prot_bin            imm_bin            rpcrate
##         4.259e-01         -1.106e-01          1.536e-01          4.370e-01
##           prorate        Age:imm_bin    lastsal:imm_bin   lastrat:prot_bin
##         3.090e-02         -4.201e-03         -2.765e-05         -6.567e-04
##   zresNORD:avgsal     zresSUD:avgsal  zresNORD:recvsaff   zresSUD:recvsaff
##        -4.552e-06          6.224e-06         -6.086e-03          2.924e-03
##  recvsaff:imm_bin      zresNORD:kept       zresSUD:kept       kept:imm_bin
##        -4.893e-02          5.225e-02         -5.455e-02          1.448e-01
##   imm_bin:rpcrate   prot_bin:prorate
##         4.299e-01          4.933e-01
```

d) Build the misclassification table on the validation sample and comment on the type I and type II errors by assuming logC(1\|0)/C(0\|1) = 0.4

```
valida.score <- predict(mod.fin, newdata = validation, type = "response")
```

We generate a vector (valida.score) containing the estimated probability that each observation in the validation dataset belongs to the positive class based on *mod.fin*

Knowing that the logarithm of the cost ratio is equal to 0.4, we can derive the cut-off value using the formula: 1/(1+e^(−0.4)).

```
cut_off_value <- 1 / (1 + exp(-0.4))
flag.estimated <- valida.score > cut_off_value #We classify observations based on predicted probabilities
misclassification_table <-table(validation$Flag,flag.estimated) #It classifies each observation: TRUE if its predicted probability (valida.score) is above the cut_off_value, FALSE otherwise.

misclassification_table
```

```
##    flag.estimated
##     FALSE TRUE
##   0   462  118
##   1   220  476
```

The misclassification table shows: 462 True Negatives (TN): Correctly predicted as FALSE (0). 118 False Positives (FP): Incorrectly predicted as TRUE (1) when actually FALSE (0). 220 False Negatives

(FN): Incorrectly predicted as FALSE (0) when actually TRUE (1). 476 True Positives (TP): Correctly predicted as TRUE (1).

```
FP <- misclassification_table[1, 2]  #falsi positivi (error type II)
FN <- misclassification_table[2, 1]  #falsi negativi (error type I)

FP/nrow(dati) #error type II rate

## [1] 0.02236543

FN/nrow(dati) #error type I rate

## [1] 0.04169826
```

Approximately 2.24% of the negative cases were incorrectly classified as positive (error type II) and approximately 4.17% of the positive cases were incorrectly classified as negative (error type I) so there is a slightly higher rate of false negatives.